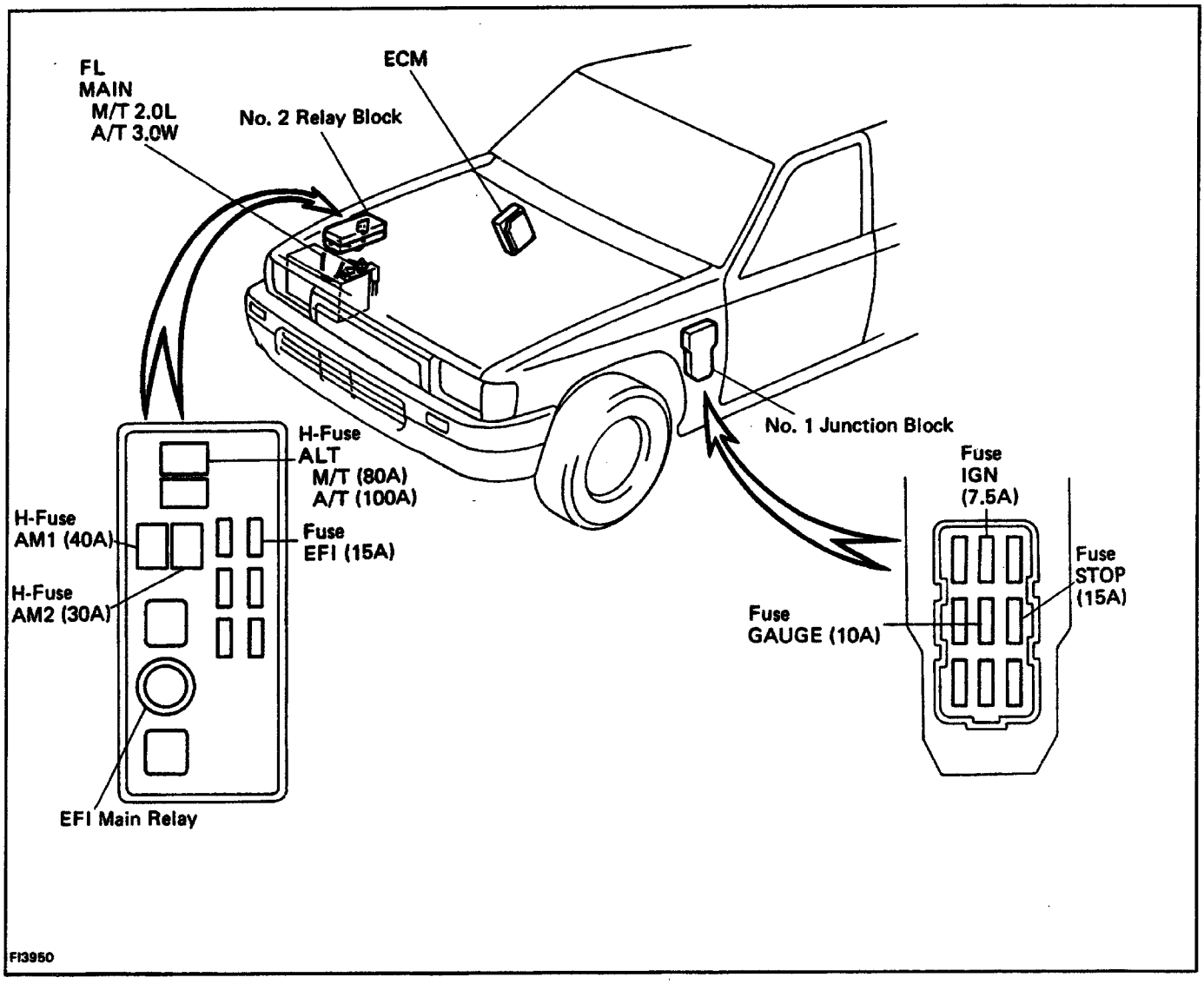
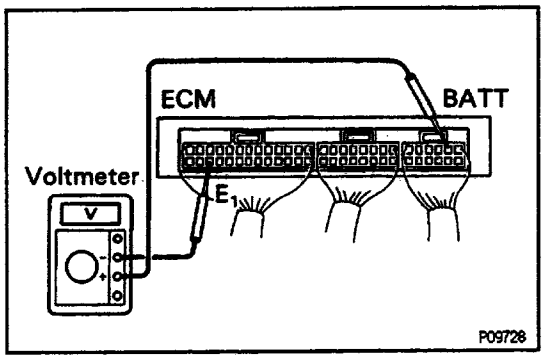
TROUBLESHOOTING w VOLT OHMMETER

HINT: Because the following troubleshooting procedures are designed for inspection of each separate system, the actual troubleshooting procedure may vary somewhat.
However, please refer to these procedures and troubleshoot, conforming to the inspection methods described.
For example, it is better to first make a simple check of the fuses, fusible links and connecting condition of the connectors before making your inspection according to the procedures listed.
The following troubleshooting procedures are based on the supposition that the trouble lies in either a short or open circuit in a component outside the computer or a short circuit within the computer. If engine trouble occurs even though proper operating voltage is detected in the computer connector, then the ECM is faulty and should be replaced.

FUSES, H– FUSES AND FUSIBLE LINK LOCATION





MFI SYSTEM CHECK PROCEDURE (Except 4WD A/T)

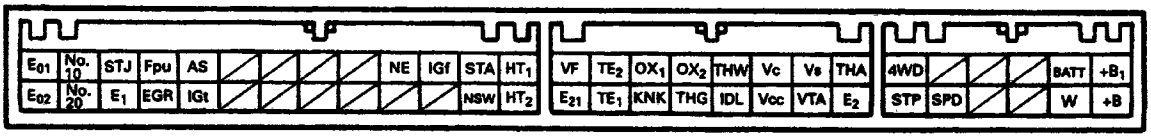
HINT:
Do all voltage measurements with the connectors connected.
Verify that the battery voltage is 11 V or more when the ignition switch is in "ON" position.
Using a voltmeter with high impedance (10 kΩ/V minimum), measure the voltage at each terminal of the wiring connector.

ECM Terminals (Except 4WD A/T)

Symbol	Terminals Name	Symbol	Terminal Name
W_{01}	ENGINE GROUND	Ox_1	OXYGEN SENSOR (MAIN)
E_{02}	ENGINE GROUND	KNK	KNOCK SENSOR
No.10	INJECTOR	*1 Ox_2	OXYGEN SENSOR (SUB)
No.20	INJECTOR	THG	EGR GAS TEMPERATURE SENSOR
STJ	COLD START INJECTOR	THW	ENGINE COOLANT TEMPERATURE SENSOR
E_1	ENGINE GROUND	IDL	THROTTLE POSITION SENSOR
Fpu	FUEL PRESSURE CONTROL VSV	Vc	VOLUME AIR FLOW METER
*1 EGR	EGR VSV	Vcc	THROTTLE POSITION SENSOR
AS	AIR VSV	Vs	VOLUME AIR FLOW METER
IGt	IGNITER	VTA	THROTTLE POSITION SENSOR
Ne	DISTRIBUTOR	THA	INTAKE AIR TEMPERATURE SENSOR
IGf	iGNITER	E_2	SENSOR GROUND
STA	STARTER SWITCH	*3 4WD	4WD SWITCH
*2 NSW	PNP SWITCH	STP	STOP LIGHT SWITCH
HT_1	OXYGEN SENSOR HEATER (MAIN)	SPD	SPEED SENSOR
*1 HT_2	OXYGEN SENSOR HEATER (SUB)	BATT	BATTERY POSITIVE VOLTAGE
VF	DLC1	W	MALFUNCTION INDICTOR LAMP
E_{21}	SENSOR GROUND	$+B_1$	MAIN RELAY
TE_2	DLC1	+B	MAIN RELAY
TE_1	DLC 1		

*1: California only *2: A/T only *3: 4WD only

ECM Terminals

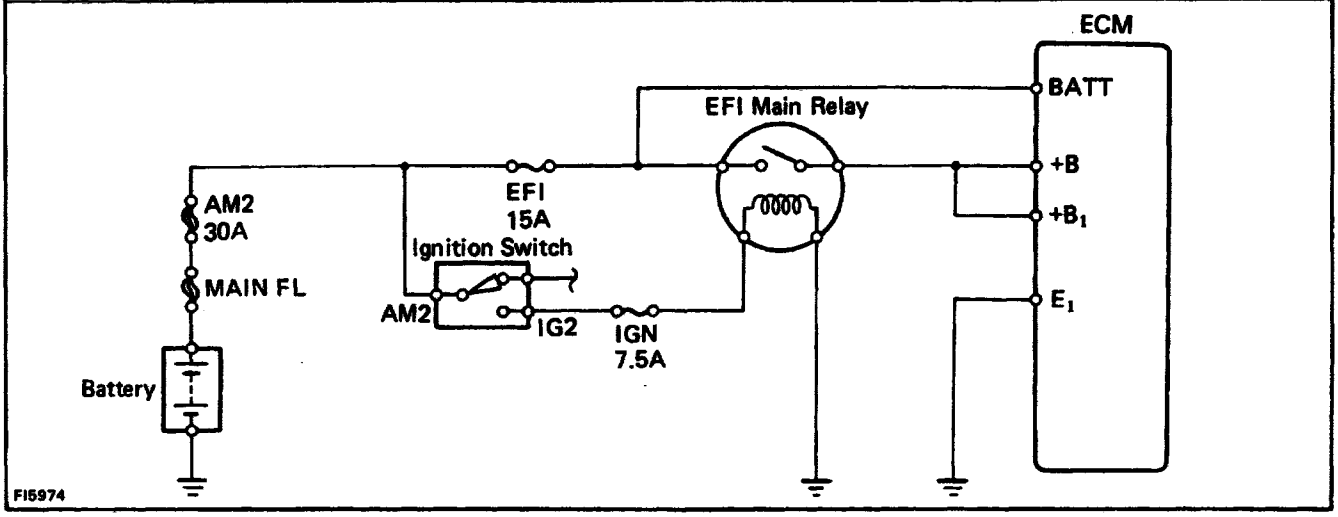


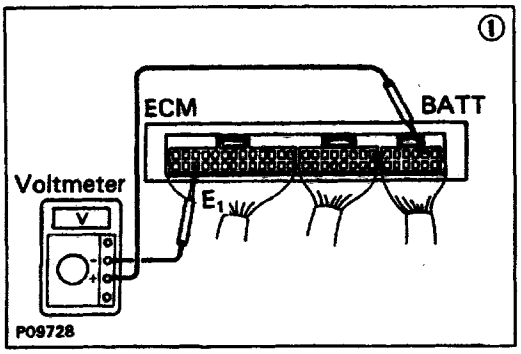
P09752

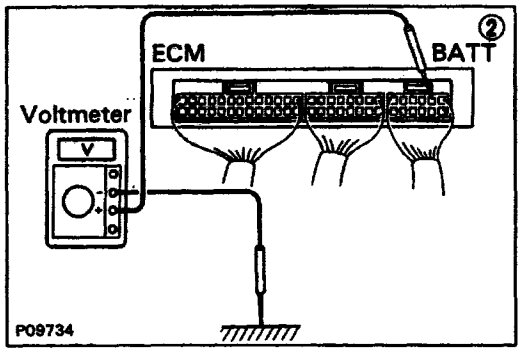
Voltage at ECM Wiring Connectors (Except 4WD A/T)

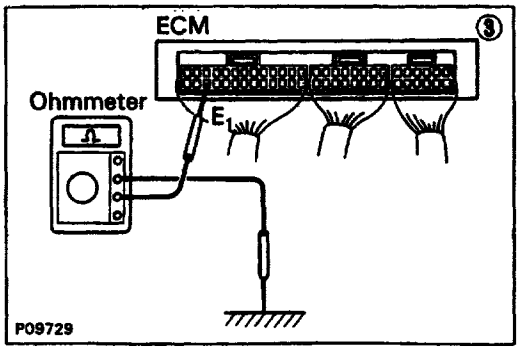
No.	Terminals	Condition		STD voltage	See page
1	BATT – E_1	–		9 – 14	EG1–128
	+B – E_1	Ignition switch ON			
	$+B_1$ – E_1				
2	IDL – E_2 (E_{21})	Ignition switch ON	Throttle valve open	9 – 14	EG1–130
	Vcc – E_2 (E_{21})		–	4.5 – 5.5	
	VTA – E_2 (E_{21})		Throttle valve fully closed	0.3 – 0.8	
			Throttle valve fully open	3.2 – 4.9	
3	Vc – E_2 (E_{21})	Ignition switch ON	–	6–10	EG1–132
	Vs – E_2 (E_{21})		Measuring plate fully closed	0.5–2.5	
			Measuring plate fully open	5–10	
		Idling		2–8	
	THA – E_2 (E_{21})	Ignition switch ON	Intake air temperature 20°C (68°F)	0.5 – 3.4	
4	THW – E_2 (E_{21})	Ignition switch ON	Coolant temperature 80°C (176°F)	0.2 – 1.0	EG1–134
5	STA – E_1	Ignition switch START position		6–12	EG1–135
6	No. 10 – E_{01} No. 20 – E_{02}	Ignition switch ON		9 – 14	EG1–136
7	IGt – E_1	Idling		0.7–1.0	EG1–137
8	W – E_1	No trouble (MIL off) and engine running		9 – 14	EG1–138
9	STJ – E_1	Ignition switch START position	Coolant temperature 80°C (176°F)	6–12	EG1–139
10	STP – E_1	Stop light switch ON		7.5 – 14	EG1–140

No.	Terminals	Trouble	Condition	STD Voltage
1	BATT – E_1	No voltage	–	9 – 14 V
	+B – E_1		Ignition switch ON	
	$+B_1$ – E_1			









• BATT–E1

(1) There is no voltage between ECM terminals BATT and Ei.

↓

Check that there is voltage between ECM terminal BATT and body ground.

NO → OK

Check wiring between ECM terminal E, and body ground.

OK → Try another ECM.

BAD → Repair or replace.

Check fuse and fusible link.

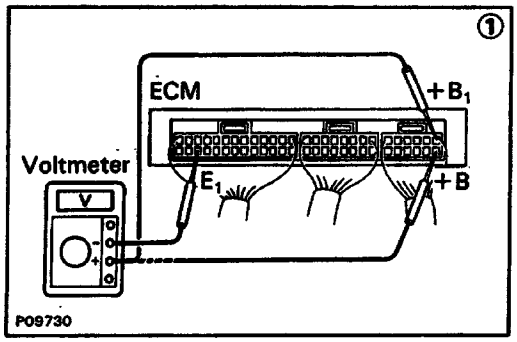
BAD → Replace.

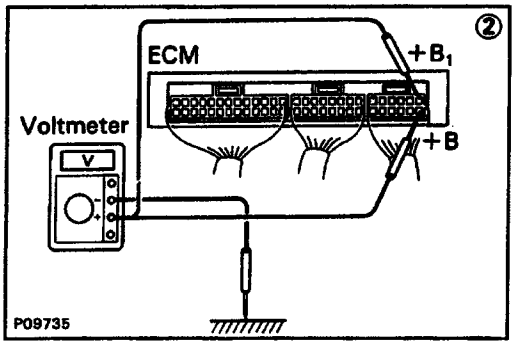
OK

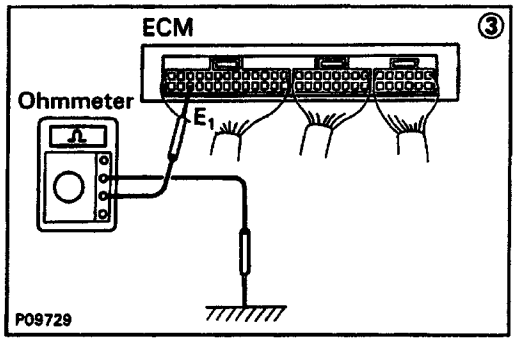
Check wiring between fuse and ECM.

BAD → Repair or replace.

V01815







- **+B (+B1) – E1**

(1) There is no voltage between ECM terminals + B (+ B1) and E1. **(IG SW ON)**

(2) Check that there is voltage between ECM terminal + B (+ B1) and body ground. **(IG SW ON)**

NO → **OK**

(3) Check wiring between ECM terminal E, and body ground.

OK → Try another ECM.

BAD → Repair or replace.

Check fuse, fusible link and ignition switch. BAD → Repair or replace.

OK

Check E F f main relay. BAD → Replace.

OK

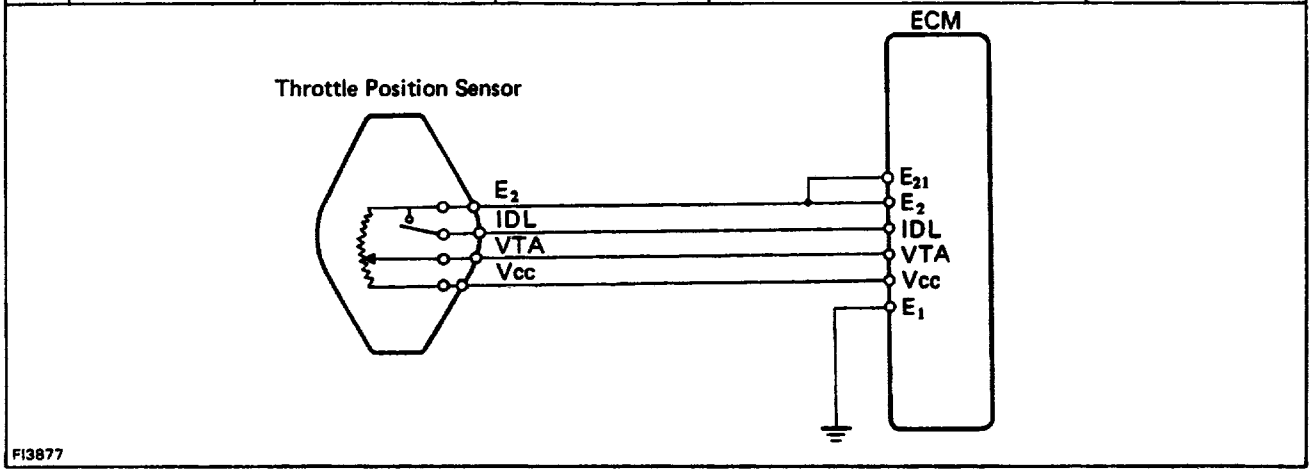
Check wiring between EFI main relay and battery. BAD → Repair or replace.

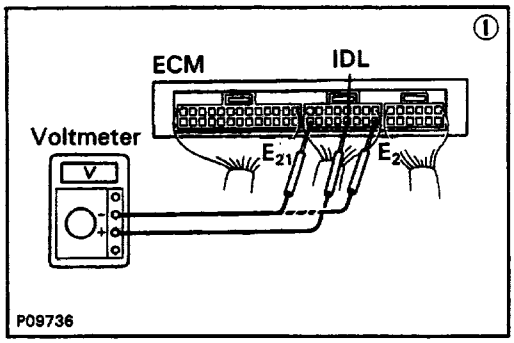
OK

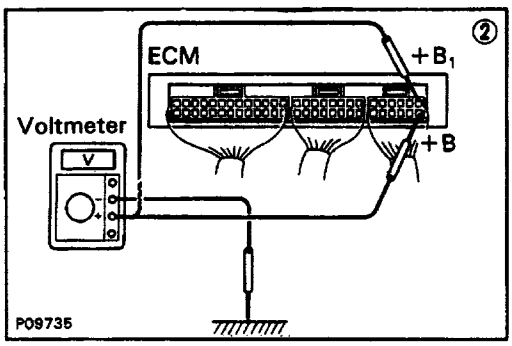
Check wiring between EFI main relay and ECM terminal +B (+B1). BAD → Repair or replace.

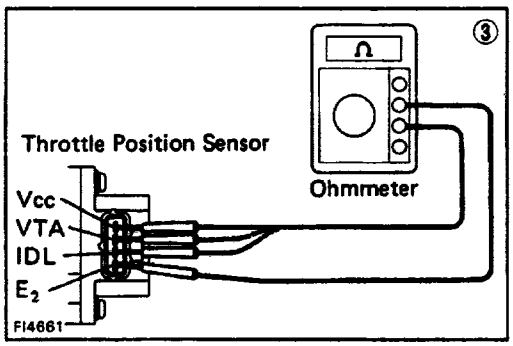
V01814

No.	Terminals	Trouble	Condition		STD Voltage
2	IDL – E_2 (E_{21})	No voltage	Ignition switch ON	Throttle valve open	9 – 14 V
	Vcc – E_2 (E_{21})			–	4.5 – 5.5 V
	VTA – E_2 (E_{21})			Throttle valve fully dosed	0.3 – 0.8 V
				Throttle valve fully open	3.2 – 4.9 V









o IDL – EZ (E21)

(1) There is no voltage between ECM terminals IDL and E2 (E2,). UG SW ON)(Throttle valve open)

(2) Check that there is voltage between ECM terminal + B (+ B1) and body ground. UG SW ON)

NO → Refer to No. 1. → BAD → Replace or repair.

Refer to No. 1. → OK → (3) Check throttle position sensor.

(3) Check throttle position sensor.

BAD → Replace or repair throttle position sensor.

OK → Check wiring between ECM and throttle position sensor.

Check wiring between ECM and throttle position sensor. → OK → Try another ECM.

OK → Check wiring between ECM terminal E, and body ground.

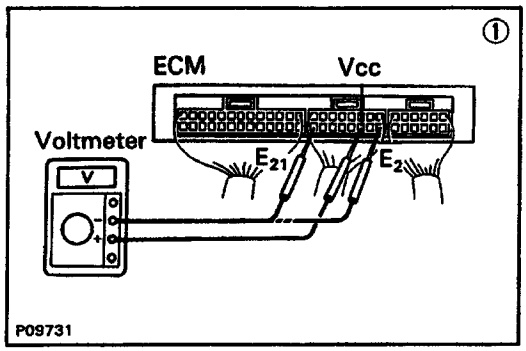
Check wiring between ECM terminal E, and body ground.

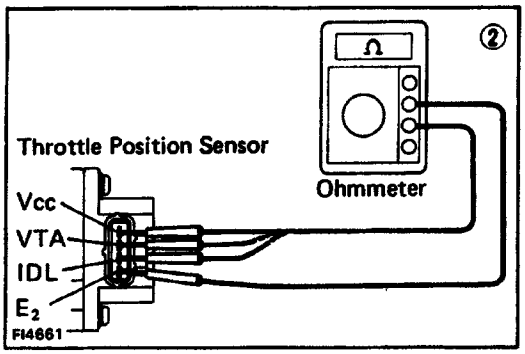
OK → (3) Check throttle position sensor.

BAD → Replace or repair.

Replace or repair. → BAD → Check wiring between ECM and throttle position sensor.

OK → (3) Check throttle position sensor.





- **Vcc – E2 (E21)**

(1) There is no voltage between ECM terminals Vcc and E2 (E21) To **(IG SW ON)**

Check that there is voltage between ECM terminals + B (+ BI) and El. UC3 SW ON)

OK → (2) Check throttle position sensor.

NO → Refer to No. t .

(2) Check throttle position sensor.

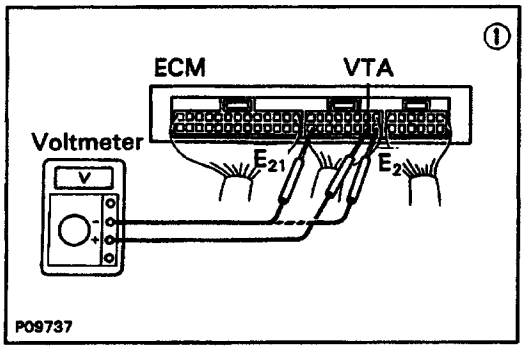
BAD → Repair or replace.

OK → Check wiring between ECM and throttle position sensor.

Check wiring between ECM and throttle position sensor.

OK → Try another ECM.

BAD → Repair or replace wiring.



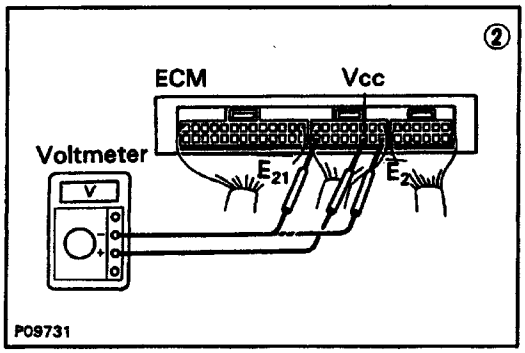
- **VTA – E2 (E21)**

(1) There is no specified voltage between ECM terminals VTA and E2 (E21). **(IG SW ON)**

(2) Check that there is voltage between ECM terminals Vcc and E2 (E21) **(IG SW ON)**

OK

NO → Perform inspection of Vcc – E2 (E21).



(3) Check throttle position sensor.

BAD → Repair or replace.

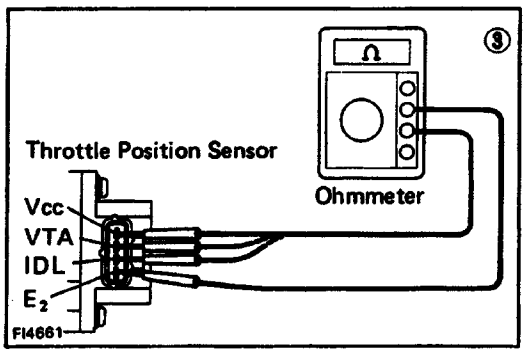
OK

Check wiring between ECM and throttle position sensor.

BAD → Repair or replace.

OK

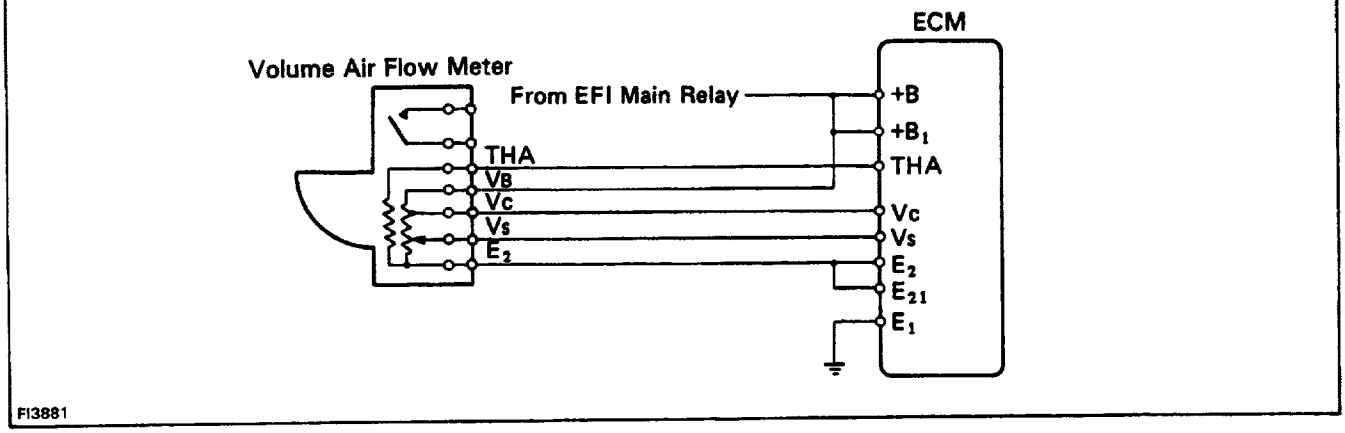
Try another ECM.

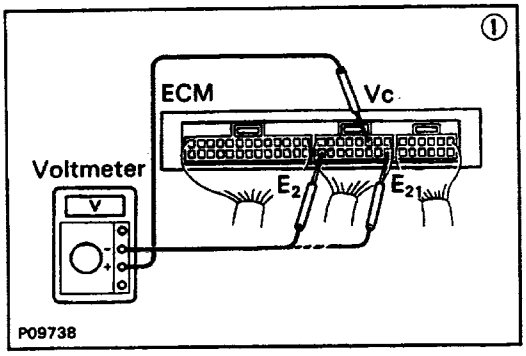


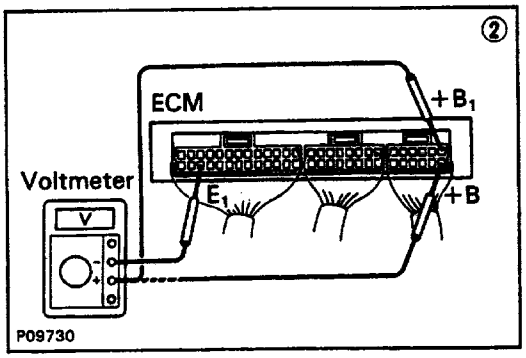
V01817

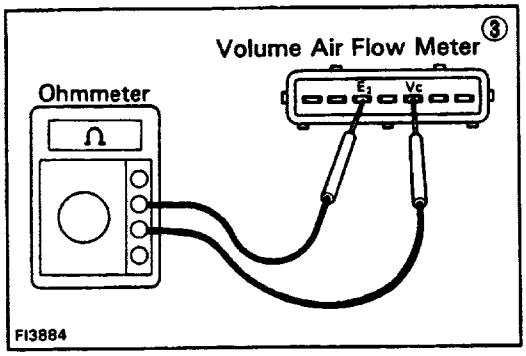
Vc –Et (E21)

No.	Terminals	Trouble	Condition		STD Voltage
3	Vc – E_2 (E_{21})	No voltage	Ignition switch ON	–	6 – 10 V
	Vs – E_2 (E_{21})			Measuring plate folly cloud	0.5 – 2.5 V
				Measuring plate fully open	5 – 10 V
			Idling		2 – 8 V
	THA – E_2 (E_{21})		ignition switch 4N	Intake air temperature 200° C (68° F)	0.5 – 3.4 V









- **Vc–E2 (E21)**

(1) There is no voltage between ECM terminals Vc and EZ (E20) **(IG SW ON)**

(2) Check that there is voltage between ECM terminals + B 1 (+ B1) and E1**. (IG SW ON)**

OK → (3) Check volume air flow meter.

NO → Refer to No. 1.

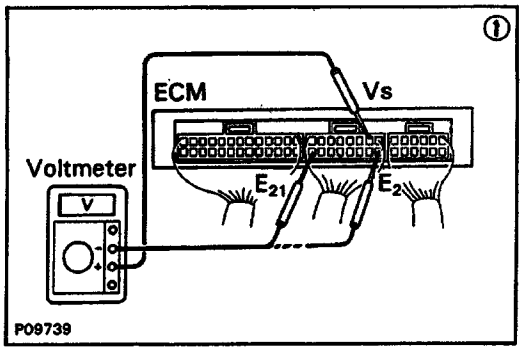
BAD → Replace or repair volume air flow meter.

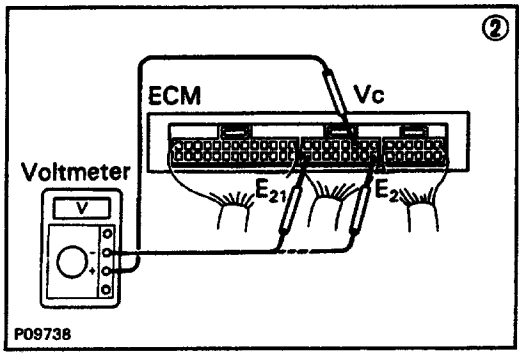
OK → Check wiring between ECM and volume air flow meter.

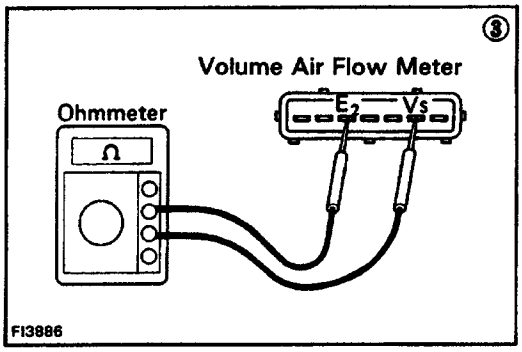
OK → Try another ECM.

BAD → Replace or repair wiring.

V01818







- **Vs – E? (E21)**

(1) There is no voltage between ECM terminals Vs and E2 (E21) **(IG SW ON)**

↓

(2) Check that there is voltage between ECM terminals Vc and E2 (E21). **(IG SW ON)**

↓ OK — NO → Refer to Vc – Et (E21) → BAD → Repair or replace.

↓ OK

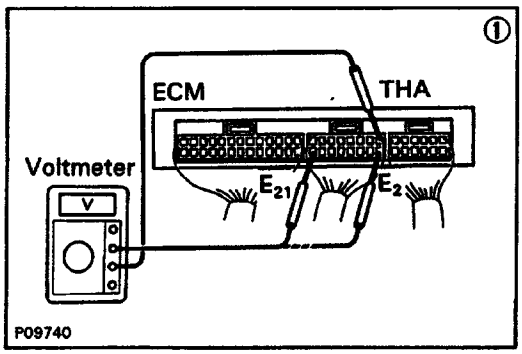
(3) Check volume air flow meter. → BAD → Repair or replace.

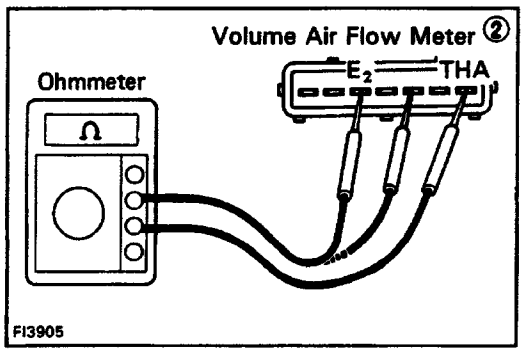
↓ OK

Check wiring between ECM and volume air flow meter. → BAD → Repair. or replace.

↓ OK

Try another ECM.





- **THA – E2 (E21)**

(1) There is no voltage between ECM terminals THA and E2 (E21) **(IG SW ON)**

↓

Check that there is voltage between ECM terminal +B (+ B1) and body ground. (IG SW ON)

↓ OK — NO → Refer to No. 1 .

↓ OK

(2) Check intake air temp. sensor.

↓ BAD → Replace volume air flow meter.

↓ OK

Check wiring between ECM and intake air temp. sensor.

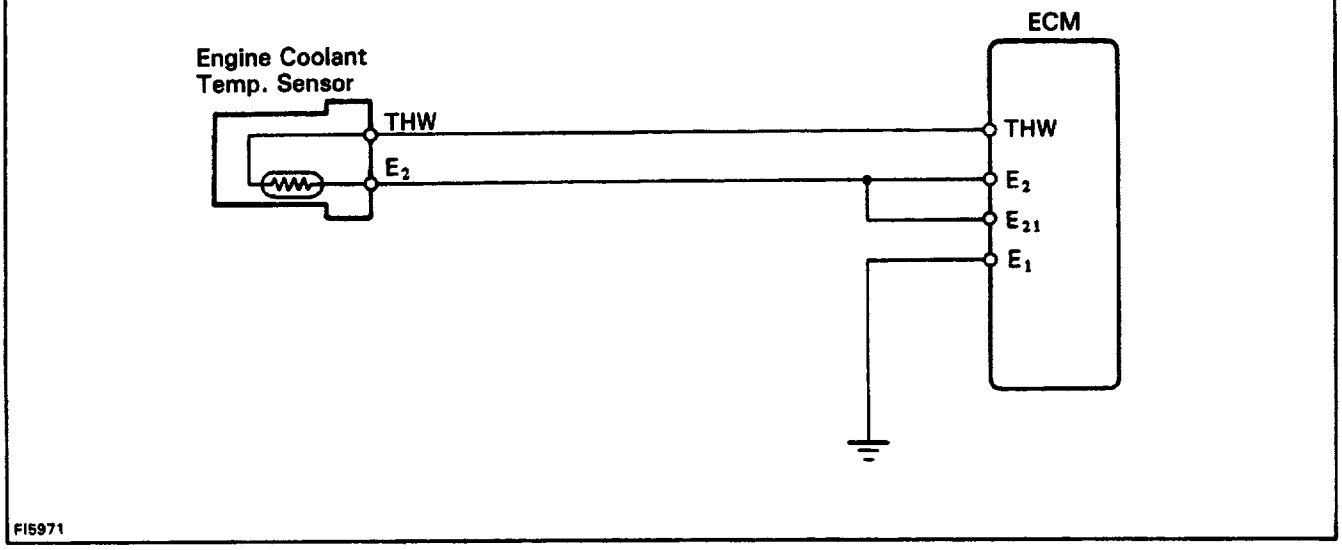
↓ OK — BAD → Repair or replace wiring.

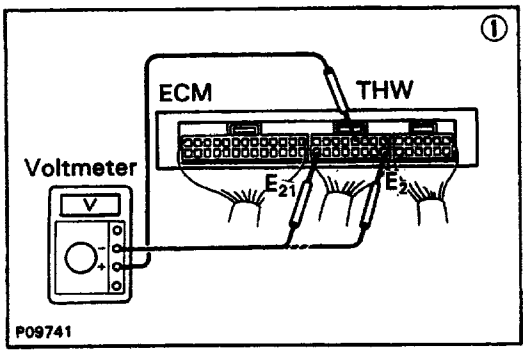
↓ OK

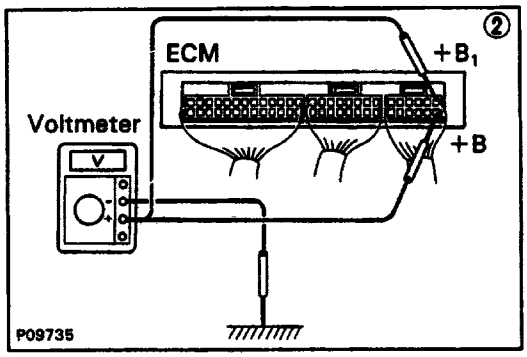
Try another ECM.

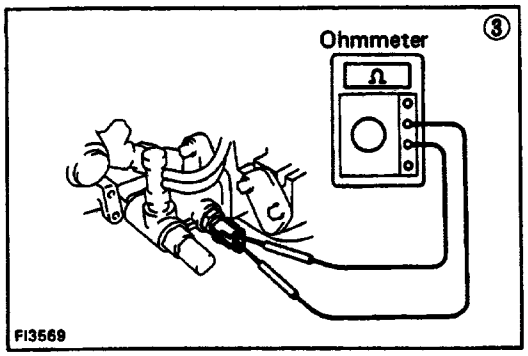
V01819

No.	Terminals	Trouble	Condition		STD –Voltage
4	THW – E_2 (E_{21})	No voltage	Ignition switch ON	Coolant temperature 80˚C (176˚ F)	0.2 – 1.0 V









(1) There is no voltage between ECM terminals THW and E2 (E20. **(IG SW ON)**

(2) Check that there is voltage between ECM terminal +B (+B) and body ground**. (IG SW ON)**

OK

NO

Refer to No. 1 .

Check wiring between ECM terminal E, and body ground.

OK

BAD

(3) Check engine coolant temp. sensor.

Repair or replace.

BAD

OK

Replace engine coolant temp. sensor.

Check wiring between ECM and engine coolant temp. sensor.

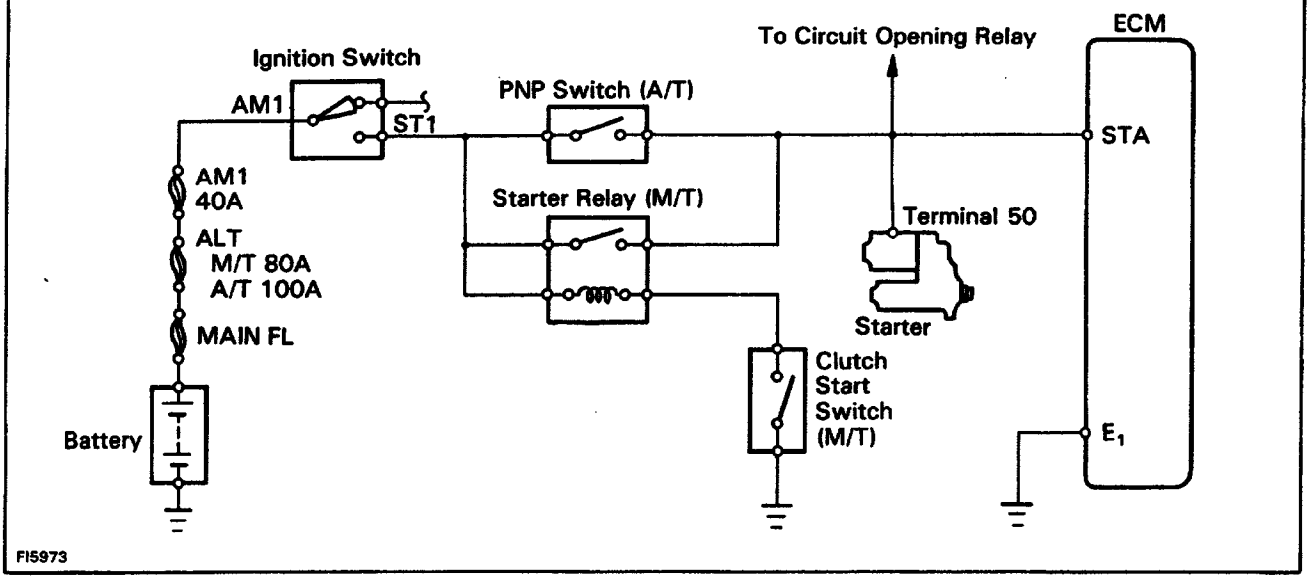
OK

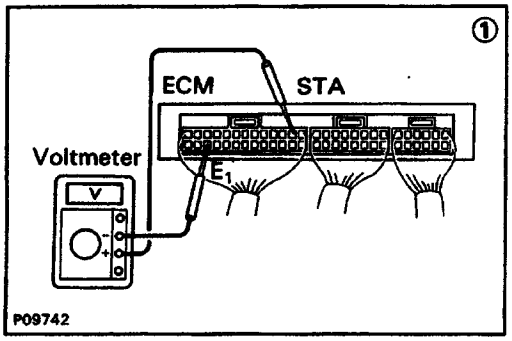
BAD

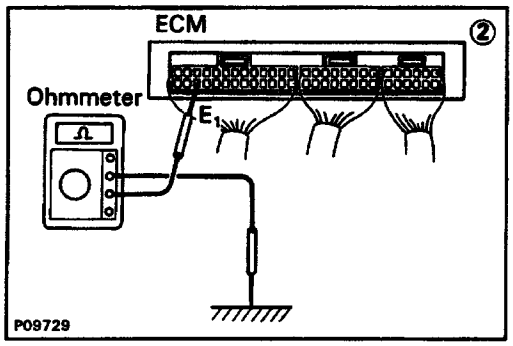
Try another ECM.

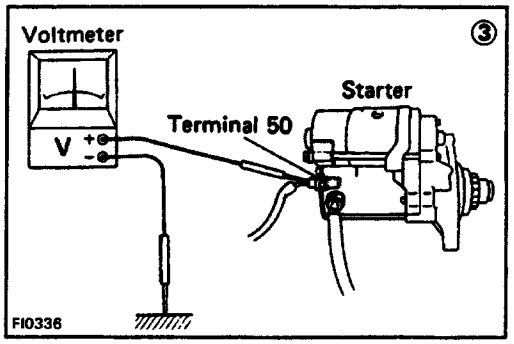
Repair or replace.

No.	Terminal:	Trouble	Condition	STD Voltage
5	STA – E_1	No voltage	Ignition switch START position	6 – 12 V









(1) There is no voltage between ECM terminals STA and EI. **(IG SW START)**

Check starter operation. — OK → Check wiring between ECM and ignition switch terminal ST.

Check starter operation. — BAD → (2) Check wiring between ECM terminal E1, and body ground.

Check wiring between ECM and ignition switch terminal ST. — BAD → Repair or replace.

Check wiring between ECM and ignition switch terminal ST. — OK → (2) Check wiring between ECM terminal E1, and body ground.

(2) Check wiring between ECM terminal E1, and body ground. — OK → Try another ECM.

(2) Check wiring between ECM terminal E1, and body ground. — BAD → Repair or replace.

Check fusible link, battery, wiring and ignition switch. — BAD → Repair or replace.

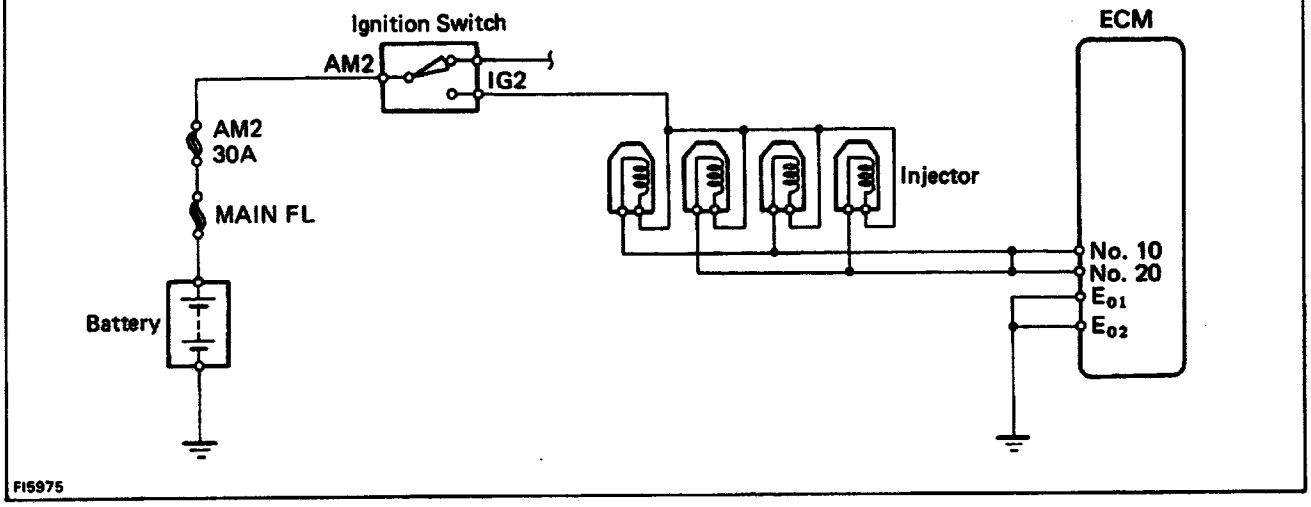
Check fusible link, battery, wiring and ignition switch. — OK → (3) Check that there is voltage at terminal 50 of starter. UG SW START) STD voltage: 6 –12 V

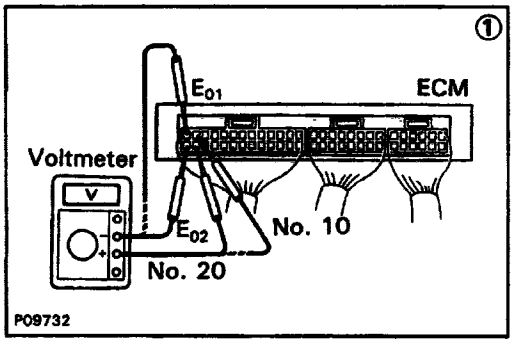
(3) Check that there is voltage at terminal 50 of starter. UG SW START) STD voltage: 6 –12 V — OK → Check starter. (See page ST–4)

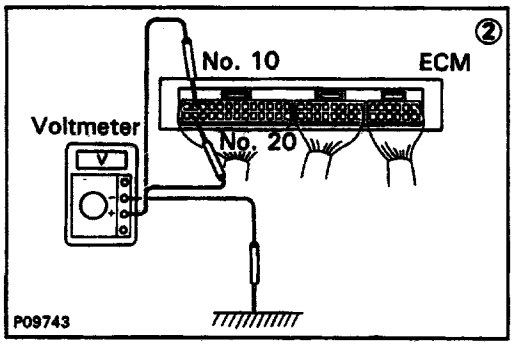
(3) Check that there is voltage at terminal 50 of starter. UG SW START) STD voltage: 6 –12 V — NO → Check wiring between ignition switch ST1 terminal and starter terminal 50.

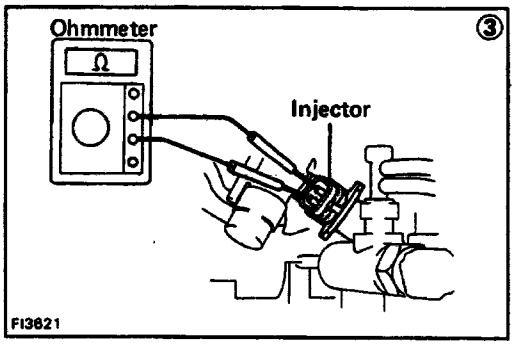
V02133

No.	Terminals	. Trouble	Condition	STD Voltage
6	No. 10 – E_{01} No. 20 E_{02}	No voltage	Ignition switch ON	9 – 14 V









(1) There is no voltage between ECM terminals No. 10 and/or No. 20 and E01 and/or E02. **(IG SW ON)**

↓

(2) Check that there is voltage between ECM terminal No. 10 and/or No. 20 and body ground.

NO → ↓

OK → Check wiring between ECM terminal E0, and/or E02 and body ground.

OK → Try another ECM.

BAD → Repair or replace.

↓

Check fusible link and ignition switch. → BAD → Repair or replace.

↓ OK

(3) Check resistance of magnetic coil in each injector.
STD resistance: 13.4 –14.2 12

OK → ↓

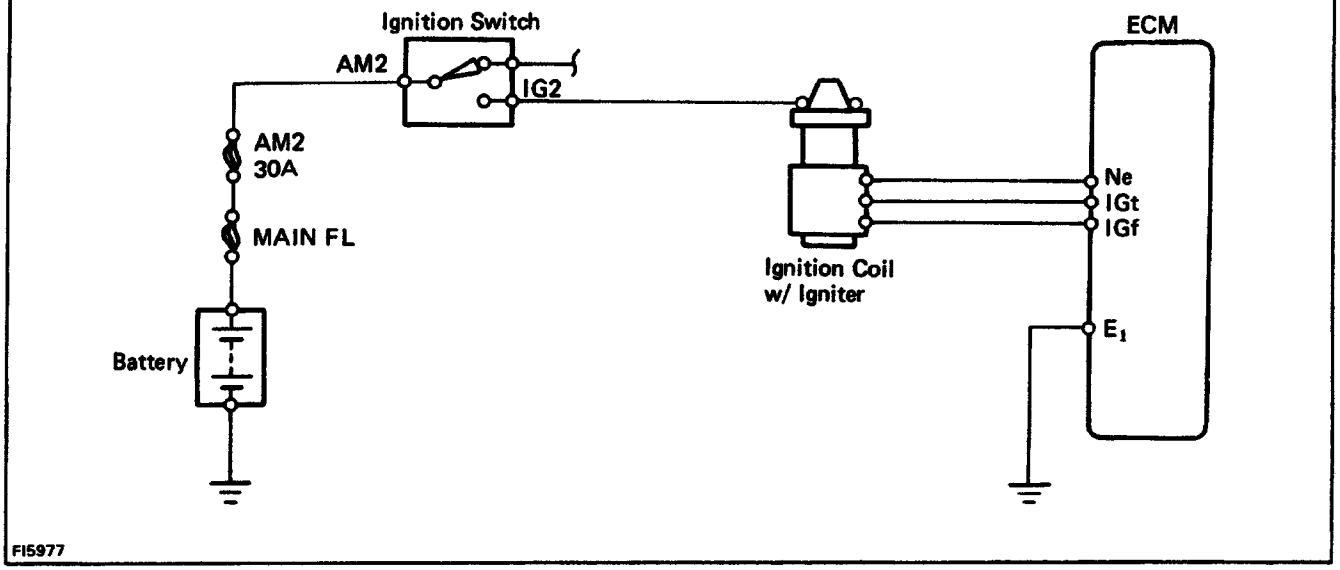
NO → Replace injector.

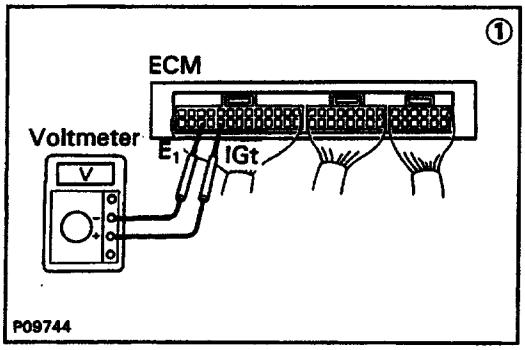
↓

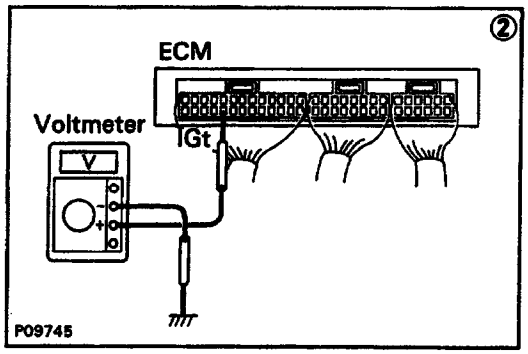
Check wiring between ECM terminal No. 10 and/or No. 20 and battery. → BAD → Repair or replace.

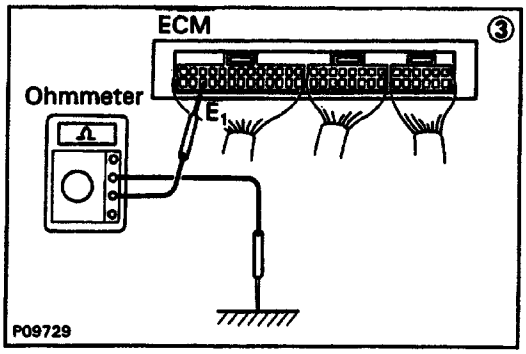
V01822

No.	Terminals	Trouble	Condition	STO Voltage
7	IGt – E_1	No voltage	Idling	0.7 – 1.0 V









(1) There is no voltage between ECM terminals IGt and E1 (Idling)

↓

(2) Check that there is voltage between ECM terminal IGt and body ground. (idling) .

NO → Refer to No. 1.

OK → (3) Check wiring between ECM terminal E1, and body ground.

BAD → Repair or replace.

Refer to No. 1. → OK

BAD → Repair or replace.

Check wiring between igniter and distributor. → OK

BAD → Replace.

Check distributor. → OK

BAD → Repair or replace.

Check wiring between ECM and igniter. → OK

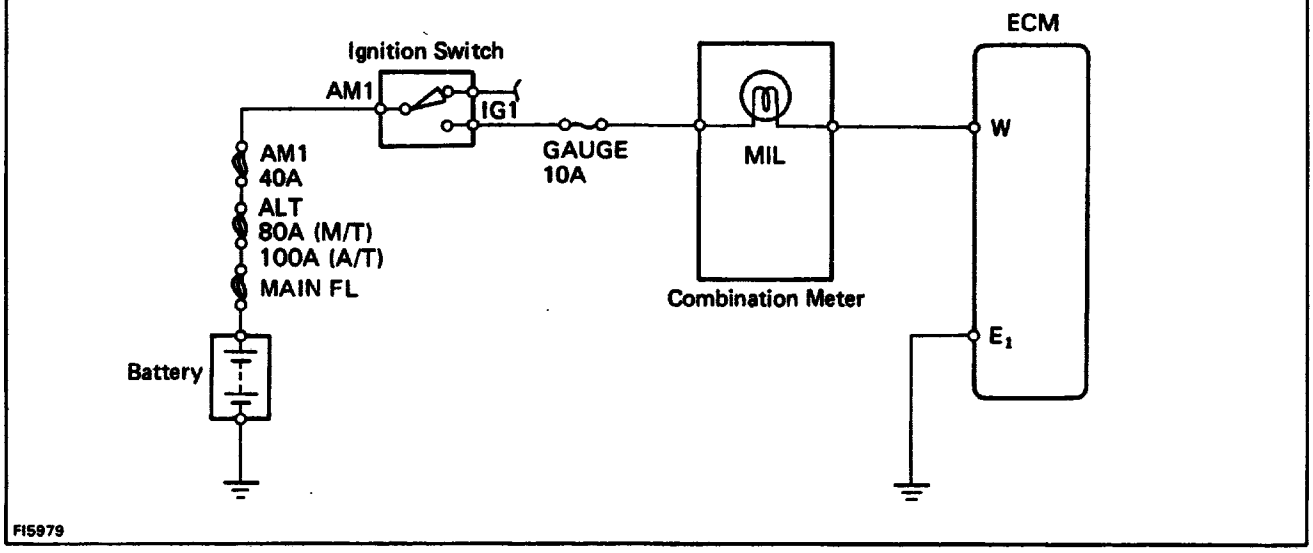
BAD → Repair or replace.

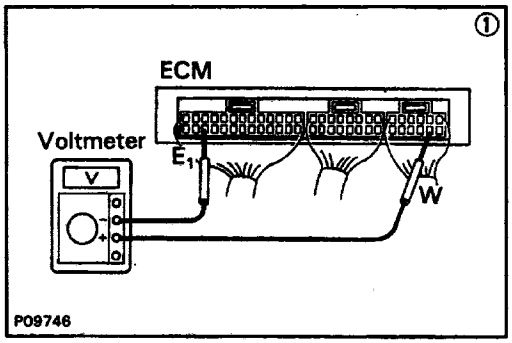
Check ignite.. → OK

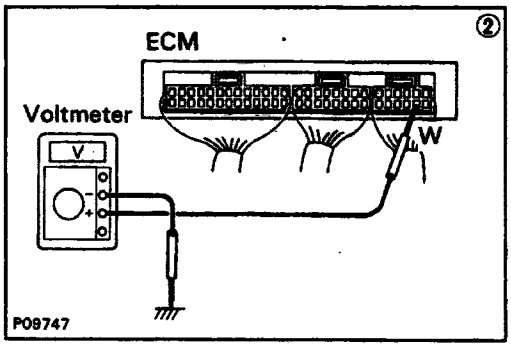
Try another ECM.

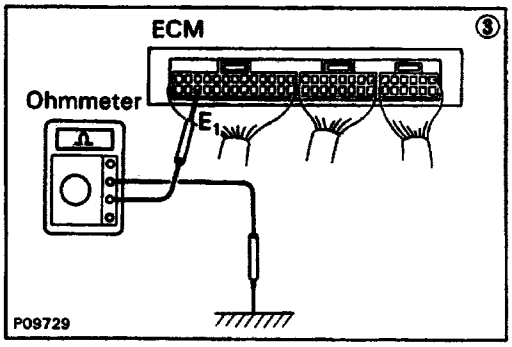
V01823

No.	Terminals	Trouble	Condition	STD Voltsp
8	W – E_1	No voltage	No trouble (MIL off) and engine running	9 – 14 V









(1) There is no voltage between ECM terminals W and El.
(Idling)

(2) Check that there is voltage between ECM terminal W and body ground.

NO

OK

(3) Check wiring between ECM terminal E1, and body ground.

OK

BAD

Try another ECM.

Repair or replace.

Check GAUGE fuse (10 Ay and MIL.

OK

BAD

Repair or replace.

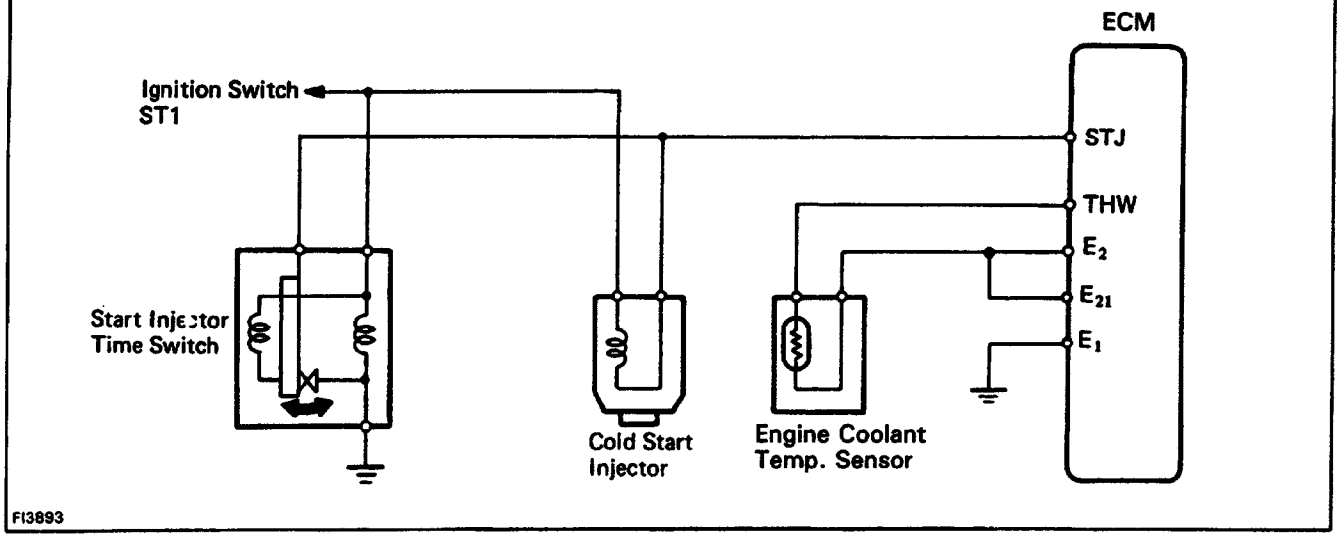
Fuse blows again
ECM BAD

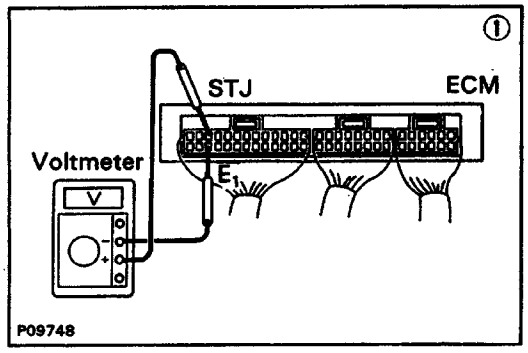
Check wiring between ECM terminal W and fuse.

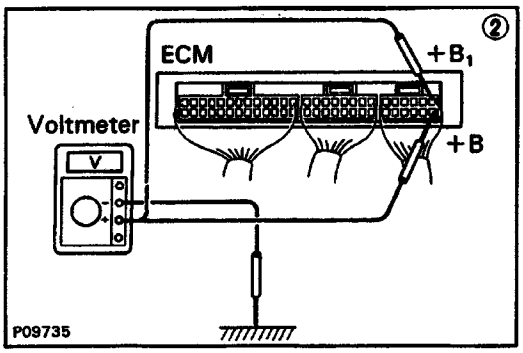
Repair or replace.

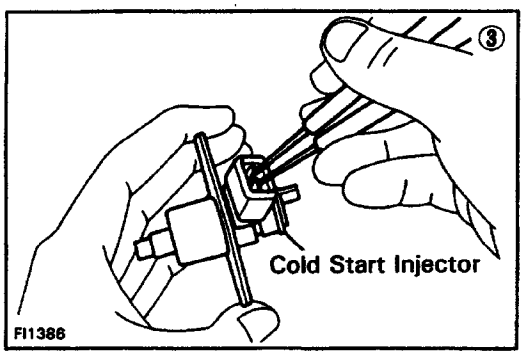
V02134

No.	Terminal	Trouble	Condition		STD Voltage
9	STJ – E_1	No voltage	Ignition switch ST position	Coolant temperature 80°C (176°F)	6 – 12 V









(1) There is no voltage between ECM terminals STJ and E1, (IG SW ON)

↓

(2) Check that there is voltage between ECM terminal + B (+B1) and body ground. **(IG SW ON)**

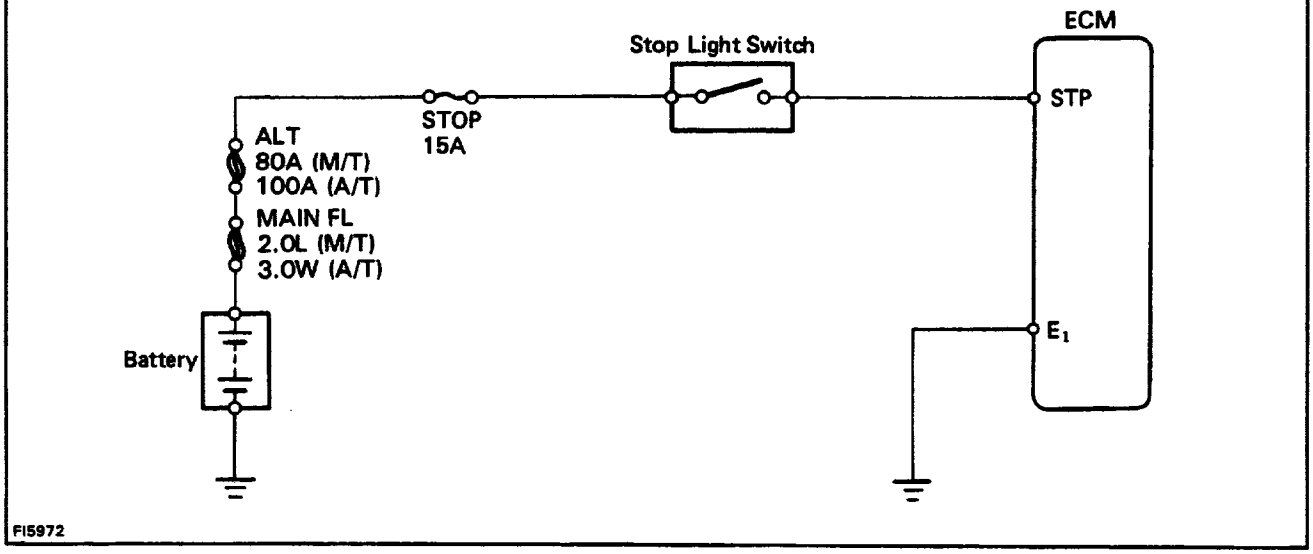
OK → (3) Check cold start injector.

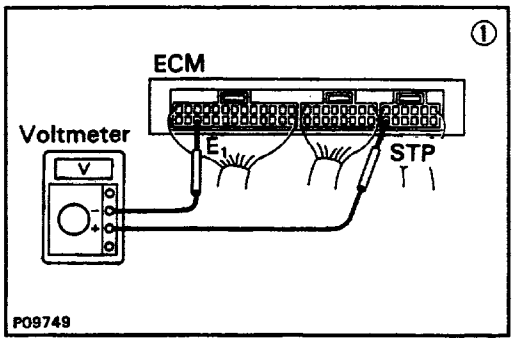
- BAD → Replace cold start injector.
- OK → Check wiring between ECM and cold start injector.
  - OK → Check wiring between ECM terminal E, and body ground.
    - OK → Try another ECM.
    - BAD → Repair or replace wiring.

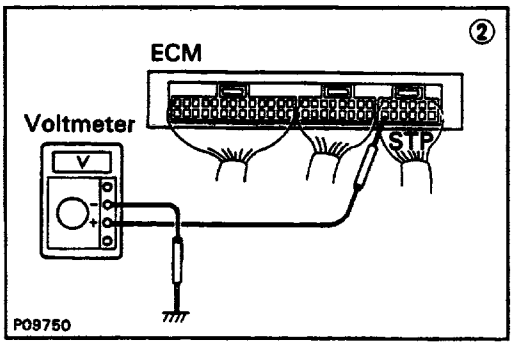
NO → Refer to No. 1.

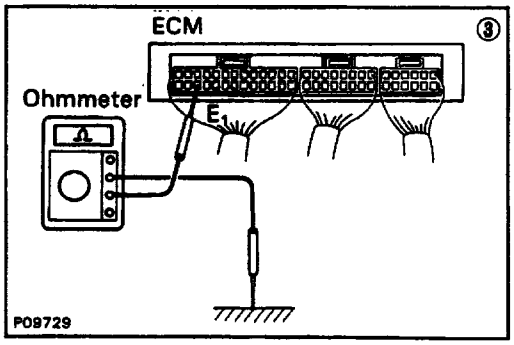
V01825

No.	Terminals	Trouble	Condition	STD Voltage
10	STP – E_1	No voltage	Stop light switch ON	7.5 – 14 V









(1)There is no voltage between ECM terminals STP and E1.

↓

(2) Check that there is voltage between ECM terminal STP and body ground when the brake pedal is depressed.

NO → ↓

OK →

(3) Check wiring between ECM terminal E, and body ground.

OK → Try another ECM.

BAD → Repair or replace.

↓

Check STOP fuse (15A) and stop light switch.

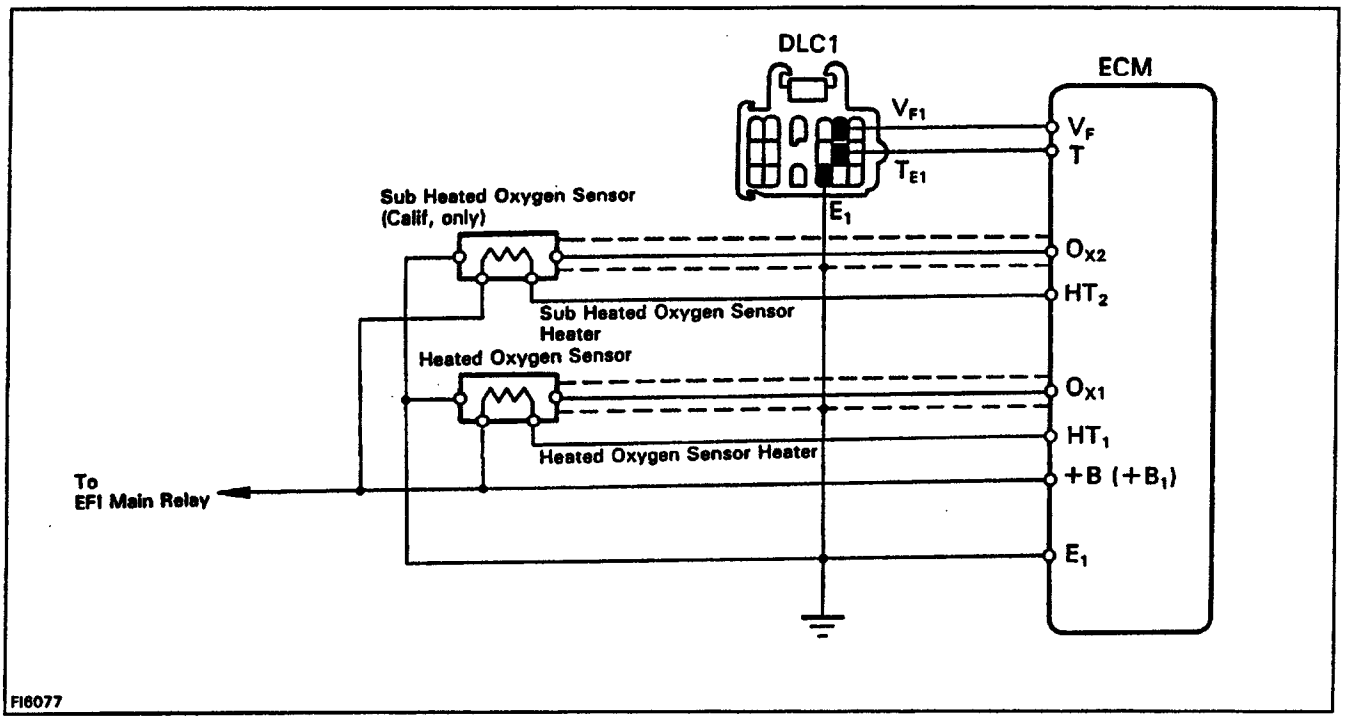
BAD → Repair or replace.

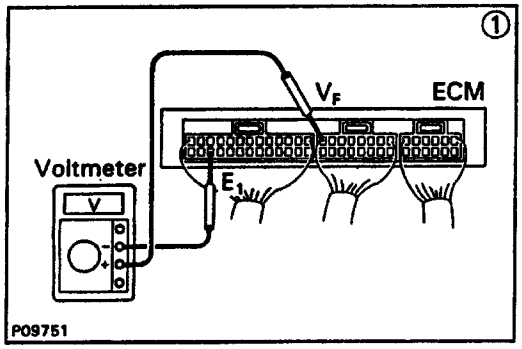
OK ↓

Check wiring between ECM terminal STP and battery.

BAD → Repair or replace.

V02135





(I)There is no voltage between ECM terminals VF and E1

Check that there is voltage between ECM terminal VF and body ground.

NO

OK

Check wiring between ECM terminal E1, and body ground.

OK

BAD

Try another ECM.

Repair or replace.

Is air leaking into air induction system?

YES

Repair air leak.

NO

Check spark plugs.

BAD

Repair or replace.

OK

Check distributor and ignition system.

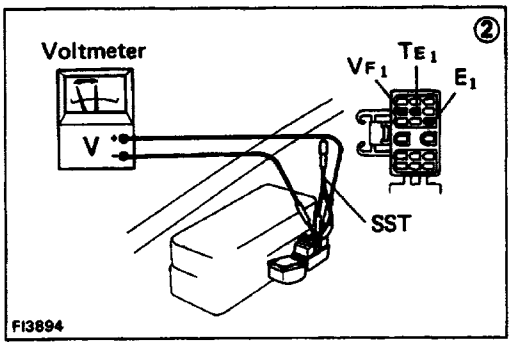
BAD

Repair or replace.

OK

CONTINUED 4N PAGE
EG1–142

V03950



CONTINUED FROM PAGE EG1–141

OK

Check fuel pressure. — BAD → Repair or replace.

OK

Check injector. — BAD → Repair or replace.

OK

Check cold start injector.* — BAD → Repair or replace.

OK

Check volume air flow meter. — BAD → Repair or replace.

OK

(2) Check operation of oxygen sensor. — OK → System normal.

BAD

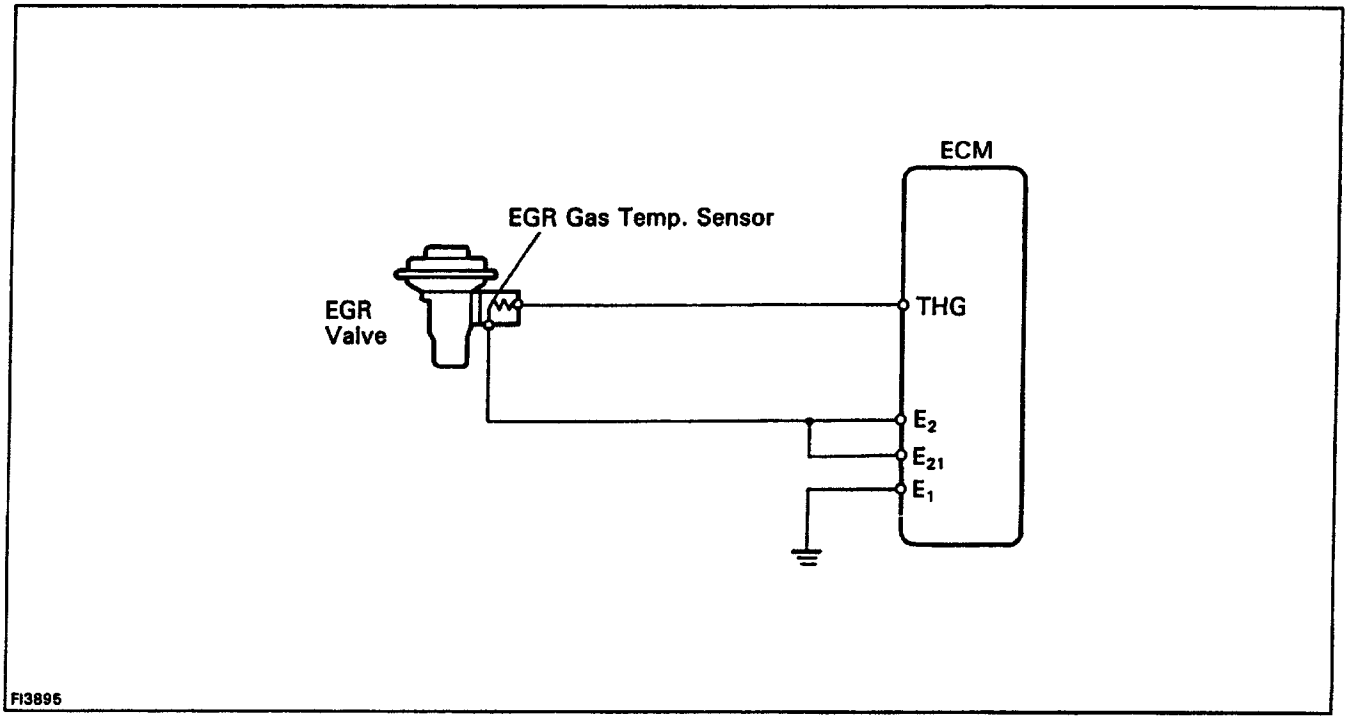
Check wiring between oxygen sensor and ECM connector. — BAD → Repair wiring.

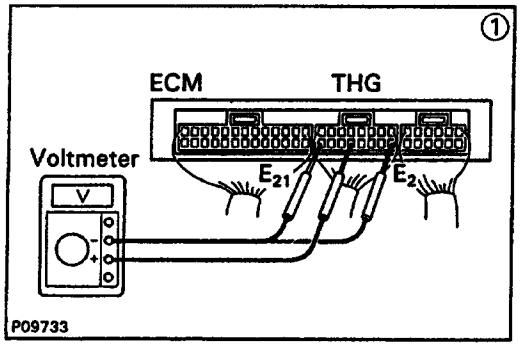
OK

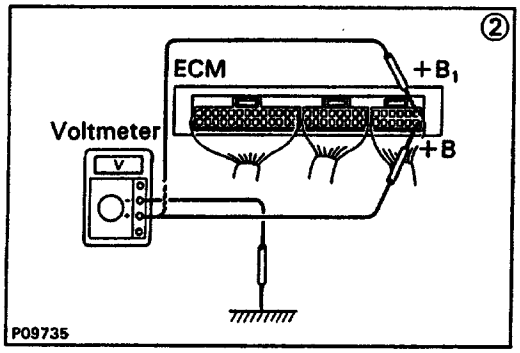
Replace oxygen sensor.

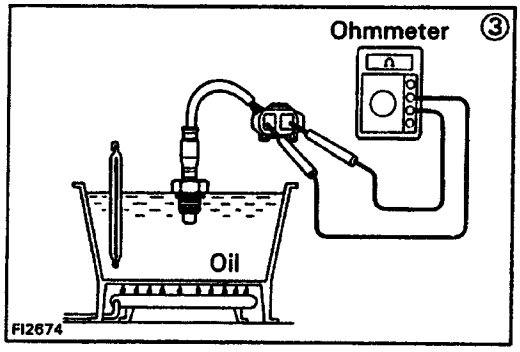
*Rich malfunction only.

V01829









(1) There is no voltage between ECM terminals THG and E2 (E21) Engine running at 2.000 rpm)

↓

(2) Check that there is voltage between ECM terminal + B (+ B1) and body ground. UG SW ON)

OK → ↓

NO → Refer to No. 1.

↓

Check wiring between ECM terminal E, and body ground.

OK → ↓

BAD → Repair or replace.

↓

Check EGR system.

BAD → Repair or replace.

OK → ↓

(3) Check EGR gas temp. sensor

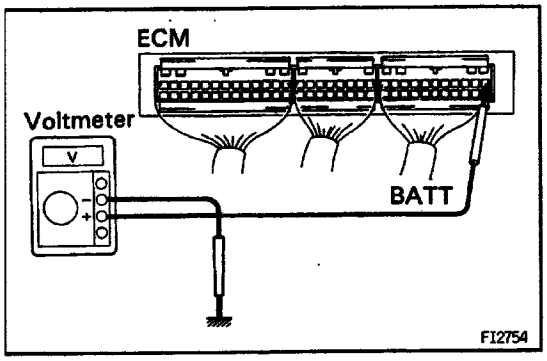
BAD → Replace EGR gas temp. sensor.

OK → Check wiring between ECM and EGR gas temp. sensor.

OK → Try another ECM.

BAD → Repair or replace.

V03951



MFI SYSTEM CHECK PROCEDURE (4WD A/T)

HINT:

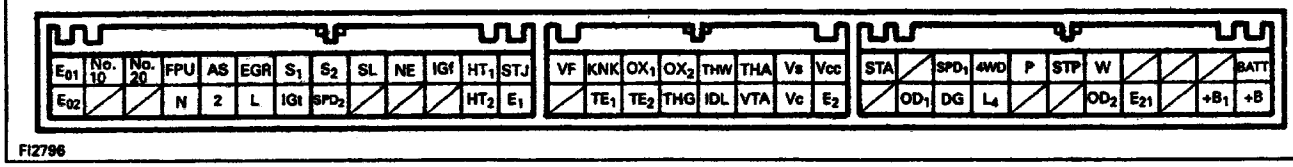
- Do all voltage measurements with the connec– tors connected.
- Verify that the battery voltage is 11 V or more when the ignition switch is in ”ON” position.
  Using a voltmeter with high impedance (10 kΩ/V minimum), measure the voltage at each terminal of the wiring connector.

ECM Terminals RWD A/T)

Symbol	Terminal Name	Symbol	Terminal Name
E_{01}	ENGINE GROUND	TE_2	DLC1
E_{02}	ENGINE GROUND	* Ox_2	OXYGEN SENSOR (SUB)
No.10	INJECTOR	THG	EGR GAS TEMPERATURE SENSOR
No.20	INJECTOR	THW	ENGINE COOLANT TEMPERATURE SENSOR
Fpu	FUEL PRESSURE CONTROL VSV	IDL	THROTTLE POSITION SENSOR
N	PNP SWITCH	THA	INTAKE AIR TEMPERATURE SENSOR
AS	PAIR VALVE	VTA	THROTTLE POSITION SENSOR
2	PNP SWITCH	Vs	VOLUME AIR FLOW METER
* EGR	EGR VSV	Vc	VOLUME AIR FLOW METER
L	PNP SWITCH	Vcc	THROTTLE POSITION SENSOR
S_1	No.1 SOLENOID	E_2	SENSOR GROUND
IGt	IGNITER	STA	SORTER SWITCH
S_2	NO..2 SOLENOID	OD_1	CRUISE CONTROL COMPUTER
SPD_2	SPEED SENSOR	SPD_1	SPEED SENSOR
S_L	SL SOLENOID	DG	DLC 1
Ne	DISTRIBUTOR	4WD	4WD SWITCH
IGf	IGNITER	L_4	TRANSFER POSITION SWITCH
HT_1	OXYGEN SENSOR HEATER (MAIN)	P	PATTERN SELECT SWITCH
* HT_2	OXYGEN SENSOR HEATER (SUB)	STP	STOP LIGHT SWITCH
STJ	COLD START INJECTOR	W	MALFUNCTION INDICATOR LAMP
E_1	ENGINE GROUND	OD_2	CRUISE CONTROL COMPUTER
V_F	DLC1	E_{21}	SENSOR GROUND
KNK	KNOCK SENSOR	$+B_1$	MAIN RELAY
TE_1	DLC 1	BATT	BATTERY POSITIVE VOLTAGE
Ox_1	OXYGEN SENSOR (MAIN)	+B	MAIN RELAY

*: California only

ECM Terminals

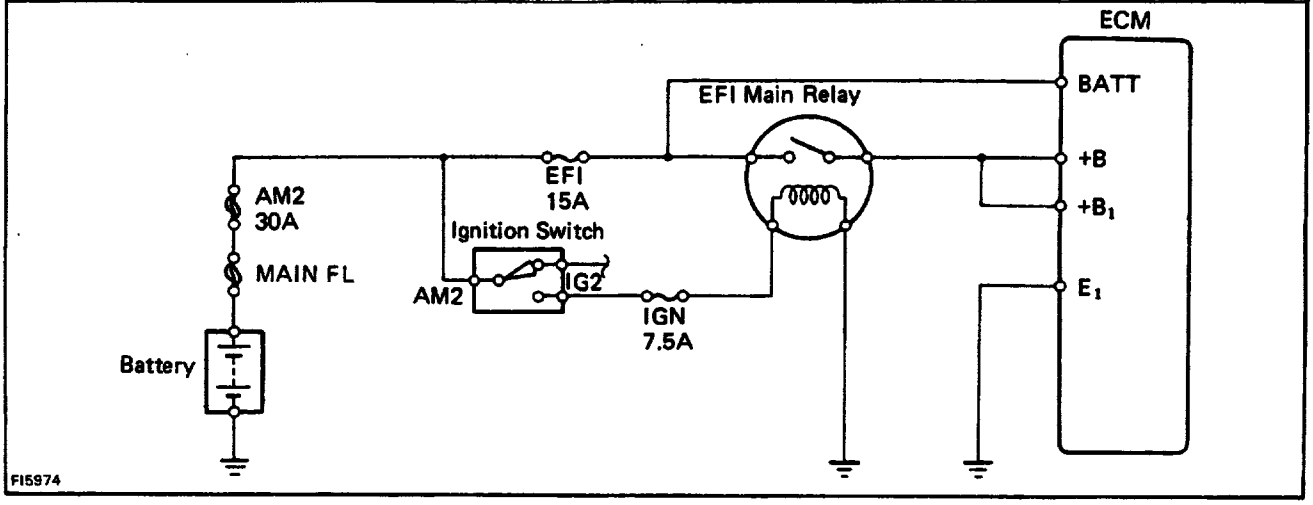


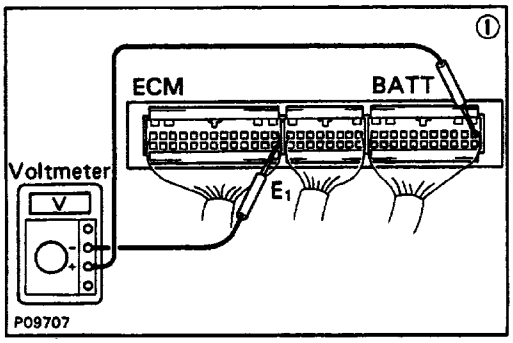
Voltage at ECM Connectors (4WD A/T)

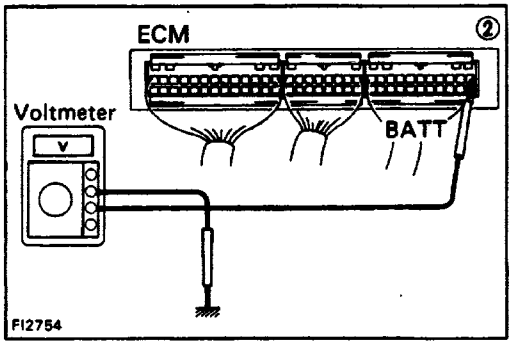
No..	Terminals	Condition		STD voltage	See page
1	BATT – E_1	–		9 – 14	EG1–146
	+B – E_1	Ignition switch ON			
	+B_1 – E_1				
2	IDL – E_2 (E_{21})	Ignition switch ON	Throttle valve open	9 – 14	EG1–148
	Vcc – E_2 (E_{21})		–	4.5 – 5.5	
	VTA – E_2 (E_{21})		Throttle valve fully closed	0.3 – 0.8	
			Throttle valve fully open	3.2 – 4.9	
3	Vc – E_2 (E_{21})	Ignition switch ON	–	6–10	EG1–150
	Vs – E_2 (E_{21})		Measuring plate fully closed	0.5 – 2.5	
			Measuring plate fully open	5 – 10	
		Idling		2 – 8	
	THA – E_2 (E_{21})	Ignition switch ON	Intake air temperature 20°C (68°F)	0.5 – 3.4	
4	THW – E_2 (E_{21})	Ignition switch ON	Coolant temperature 80°C (I76°F)	0.2 – 1.0	EG1–152
5	STA – E_1	Ignition switch START position		6–12	EG1–153
6	No. 10 – E_{01} No. 20 – E_{02}	Ignition switch ON		9 – 14	EG1–154
7	IGt – E_1	Idling		0.7–1.0	EG1–155
8	W – E_1	No trouble (MIL off) and engine running		9 – 14	EG1–156
9	STJ – E_1	Ignition switch START position	Coolant temperature 80°C (176°F)	6–12	EG1–157
10	STP – E_1	Stop light switch ON		7.5 – 14	EG1–158

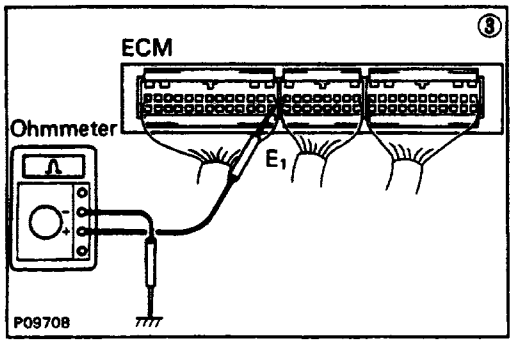
V01831

No.	Terminals	Trouble	Condition	STD voltage
1	BATT – E_1	No voltage	–	9 – 14 V
	+B – E_1		Ignition switch ON	
	$+B_1$ – E_1			









• **BATT – E 1**

(1) There is no voltage between ECM terminals BATT and E1.

↓

(2) Check that there is voltage between ECM terminal BATT and body ground.

NO → (3) Check wiring between ECM terminal E1 and body ground.

(3) OK → Try another ECM.

(3) BAD → Repair or replace.

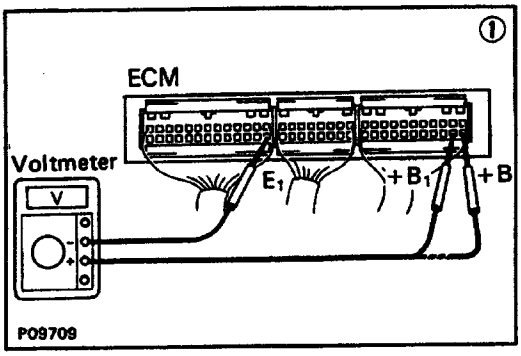
(2) OK → Check fuse and fusible link.

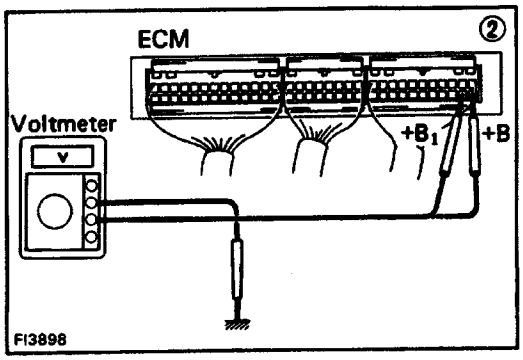
Check fuse and fusible link. BAD → Replace.

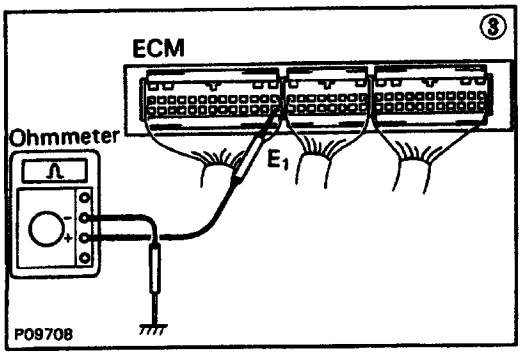
Check fuse and fusible link. OK → Check wiring between fuse and ECM.

Check wiring between fuse and ECM. BAD → Repair or replace.

V01833







- **+B (+B,) – E 1**

(1) There is no voltage between ECM terminals + B (+ B1) and E1 **(IG SW ON)**

(2) Check that there is voltage between ECM terminal +B (+B1) and body ground. **(IG SW ON)**

NO → OK

(3) Check wiring between ECM terminal E1 and body ground.

OK → Try another ECM.

BAD → Repair or replace.

Check fuse, fusible link and ignition switch. BAD → Repair or replace.

OK

Check EF I main relay. BAD → Replace.

OK

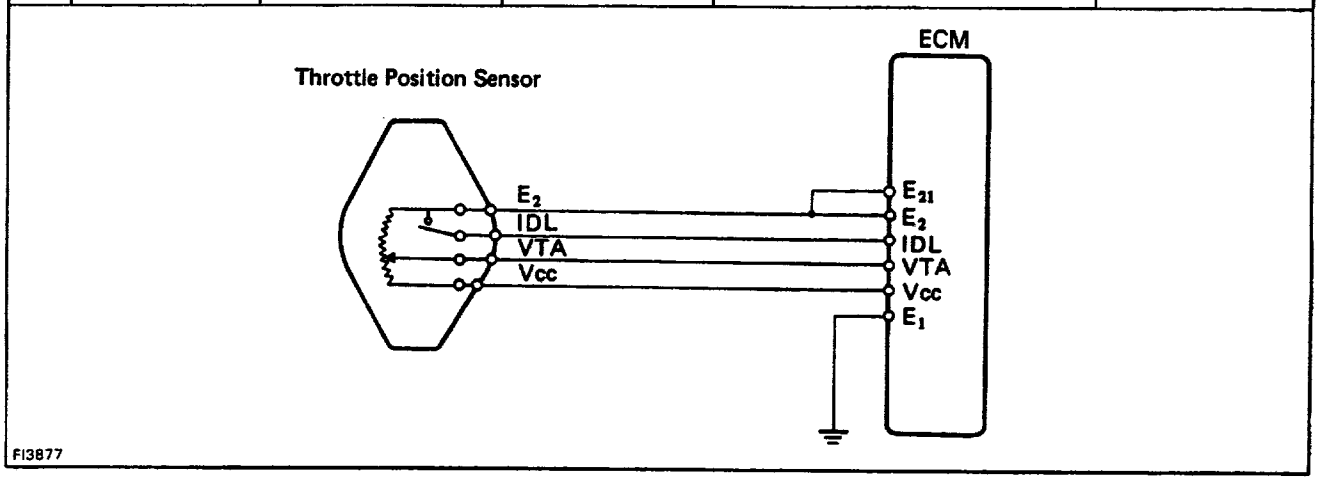
Check wiring between EF I main relay and battery. BAD → Repair or replace.

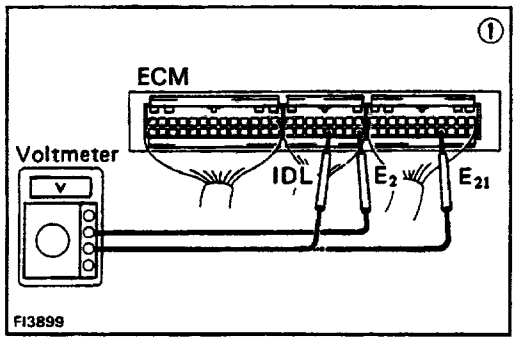
OK

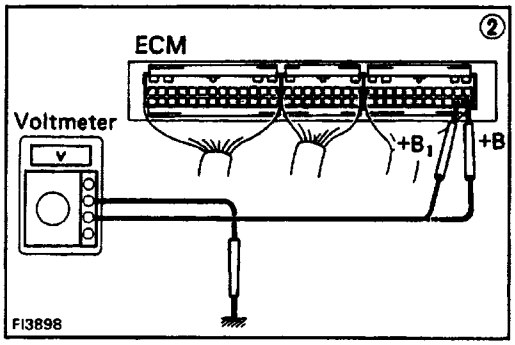
Check wiring between EFI main relay and ECM terminal +B (B+1). BAD → Repair or replace.

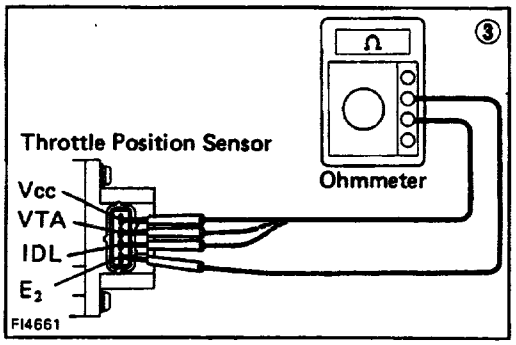
V01832

No.	Terminals	Trouble	Condition		STD Voltage
2	IDL – E_2 (E_{21})	No voltage	Ignition switch ON	Throttle valve open	9 – 14 V
	Vcc – E_2 (E_{21})			–	4.5 – 5.5 V
	VTA – E_2 (E_{21})			Throttle valve fully dosed	0.3 – 0.8 V
				Throttle valve fully open	3.2 – 4.9 V









• **IDL – EZ (E21)**

(1) There is no voltage between ECM terminals IDL and E2 (E21). **(IG SW ON)** (Throttle valve open)

(2) Check that there is voltage between ECM terminal + B (+B1) and body ground**. (IG SW ON)**

NO

Refer to No. 1. — BAD → Replace or repair.

OK

OK

Check wiring between ECM terminal E, and body ground.

OK

BAD

Replace or repair.

BAD

(3) Check throttle position sensor.

BAD

OK

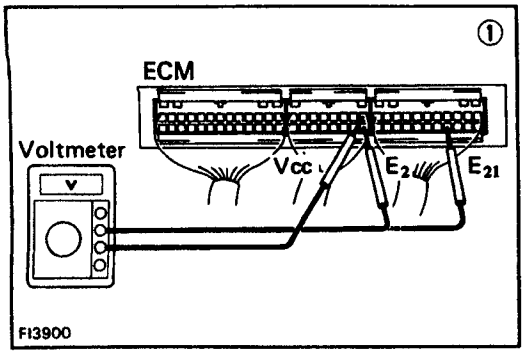
Replace or repair throttle position sensor.

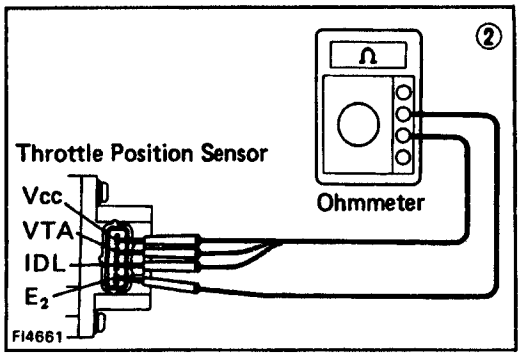
Check wiring between ECM and throttle position sensor.

OK

Try another ECM.

V01834





• **FCC – EZ (Et1)**

(1) There is no voltage between ECM terminals FCC and E2. **(IG SW ON)**

Check that there is voltage between ECM terminals +B (+ B1) and E1 **(IG SW ON)**

OK → (2) Check throttle position sensor.

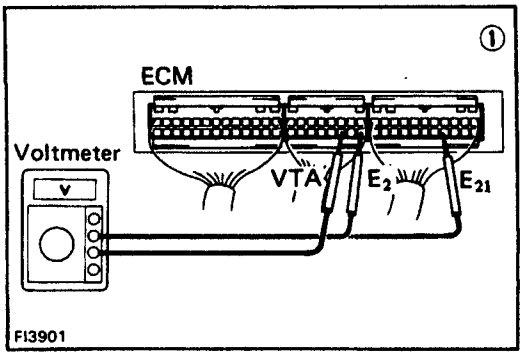
NO → Refer to No. 11.

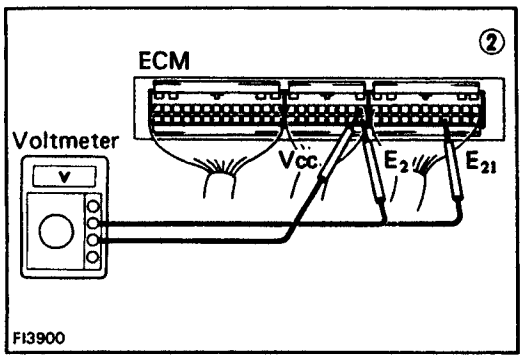
BAD → Repair or replace.

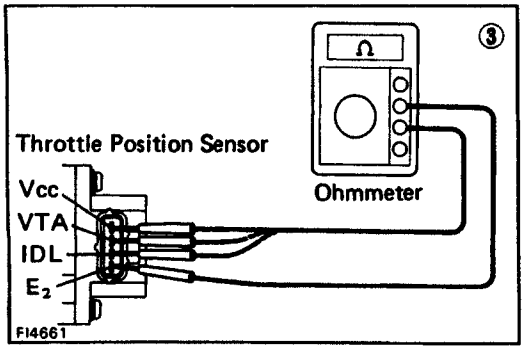
OK → Check wiring between ECM and throttle position sensor.

OK → Try another ECM.

BAD → Repair or replace wiring.







• **VTA – E2 (E21)**

(1) There is no voltage between ECM terminals VTA and E2 (E20. (IG SW ON)

(2) Check that there is voltage between ECM terminals FCC and E2 (E20. **(IG SW ON)**

OK → (3) Check throttle position sensor.

NO → Perform inspection of FCC – Et (E21).

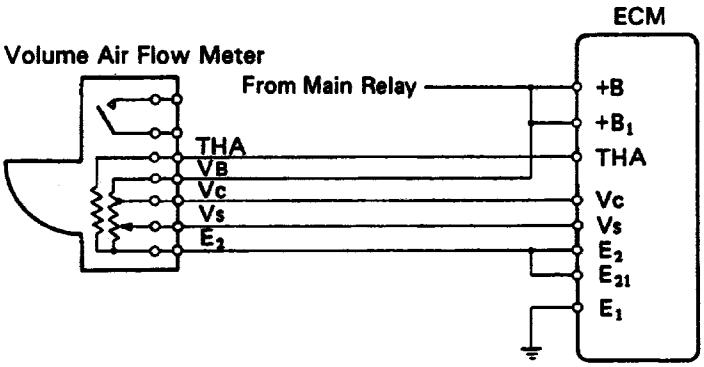
BAD → Repair or replace.

OK → Check wiring between ECM and throttle position sensor.

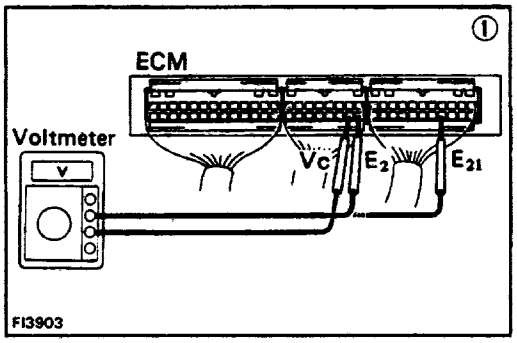
BAD → Repair or replace.

OK → Try another ECM.

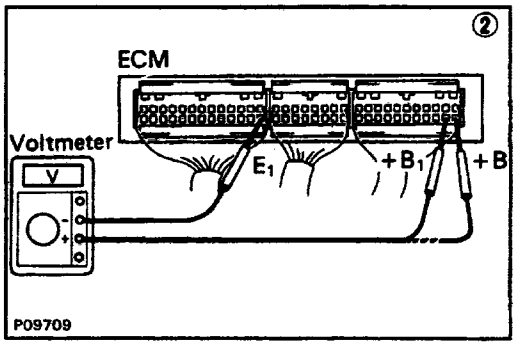
No.	Terminal:	Trouble	Condition		STD Voltage
3	Vc – E_2 (E_{21})	No voltage	Ignition switch ON	–	6 – 10 V
	Vs – E_2 (E_{21})			Measuring plate fully dosed	0.5 – 2.5 V
				Measuring plate fully open	5 – 10 V
			Idling		2 – 8 V
	THA – E_2 (E_{21})		Ignition –switch ON	Intake air temperature 20˚C (68˚ F)	0.5 – 3.4 V



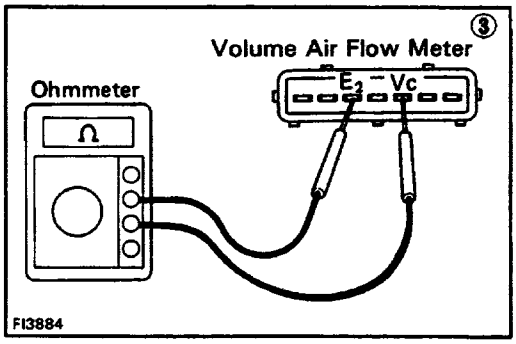
FI3881



FI3903



P09709



FI3884

• **Vc– E2 (E21)**

(1) There is no voltage between ECM terminals Vc and EZ (E21) **(IG SW ON)**

(2) Check that there is voltage between ECM terminals + B (+ B1) and E1. **(IG SW ON)**

OK — **NO**

(3) Check volume air flow meter. — Refer to No. t .

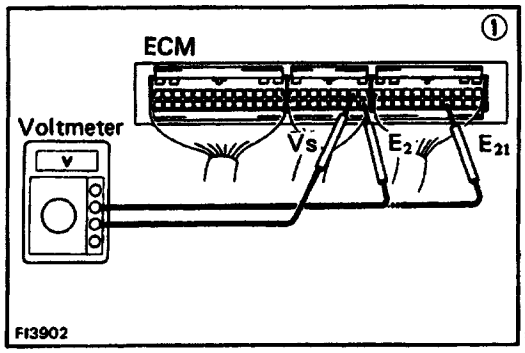
BAD — **OK**

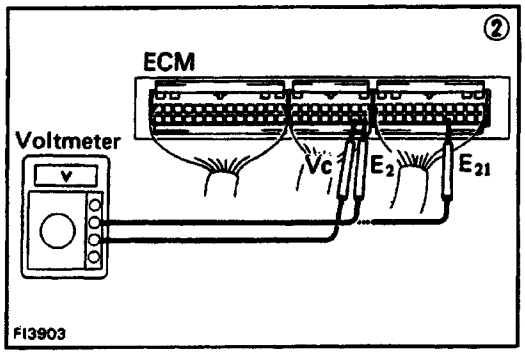
Replace or repair volume air flow meter. — Check wiring between ECM and volume air flow meter.

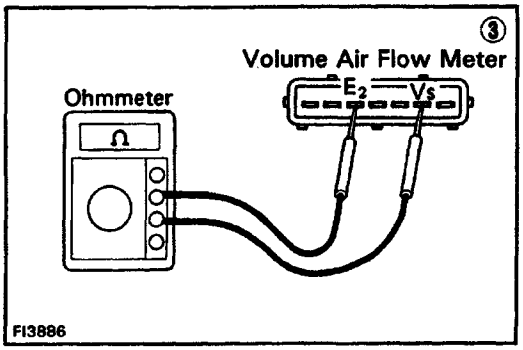
OK — BAD

Try another ECM. — Replace or repair wiring.

V01836







- **Vs – E2 (E21)**

(1) There is no voltage between ECM terminals Vs and E2 (E21) **(IG SW ON)**

↓

(2) Check that there is voltage between ECM terminals Vc and E2 (E21). **(IG SW ON)**

↓ **OK** / **NO** → Refer to FCC – E2 (E21). → BAD → Repair or replace.

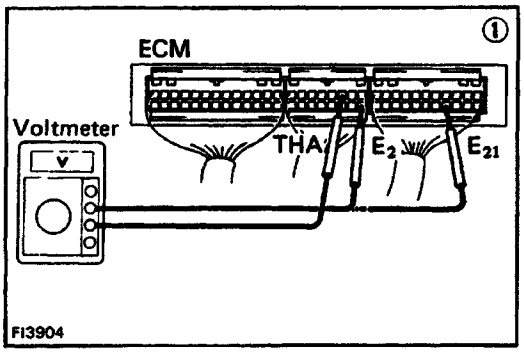
↓ (3) Check volume air flow meter. → BAD → Repair or replace.

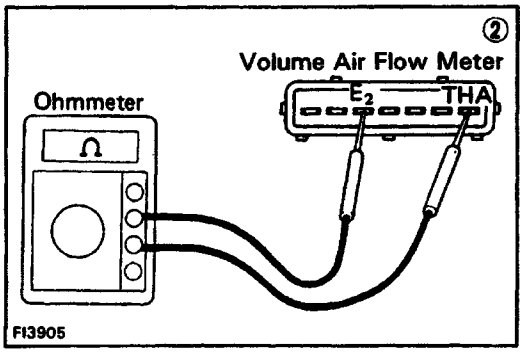
↓ **OK**

↓ Check wiring between ECM and volume air flow meter. → BAD → Repair or replace.

↓ **OK**

Try another ECM.





- **THA – E2 (E21)**

(1) There is no voltage between ECM terminals THA and E2 (E21). **(IG SW ON)**

↓

Check that there is voltage between ECM terminal + B (+ B1) and body ground **(IG SW ON)**

↓ **OK** / **NO** → Refer to No. 1.

↓ (2) Check intake air temp. sensor.

↓ BAD → Replace volume air flow meter.

↓ **OK** → Check wiring between ECM and intake air temp. sensor.

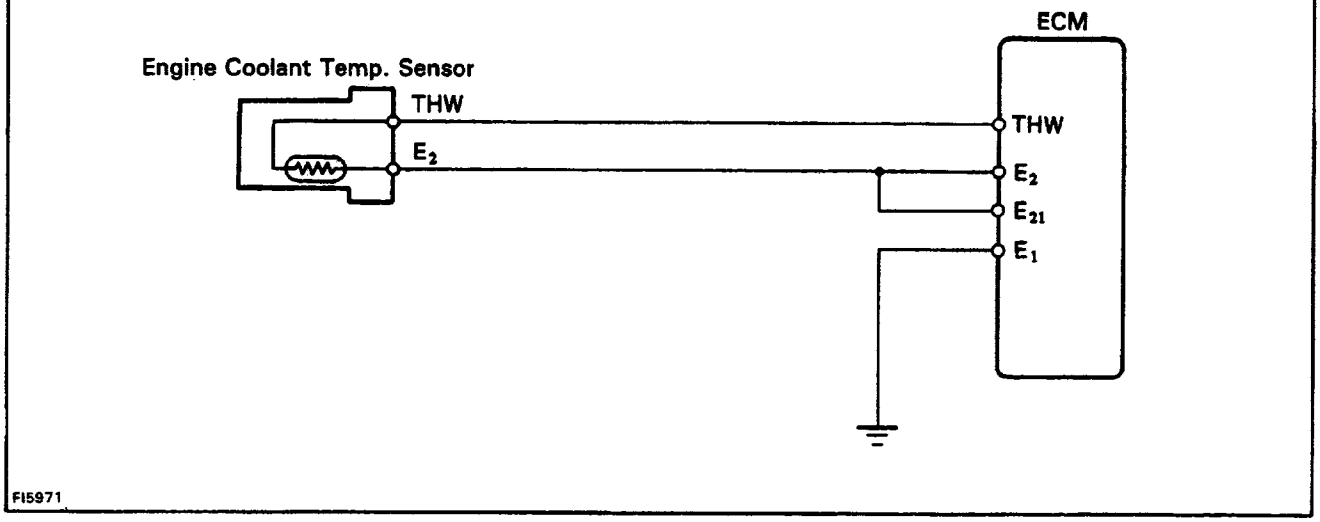
↓ **OK** / BAD → Repair or replace wiring.

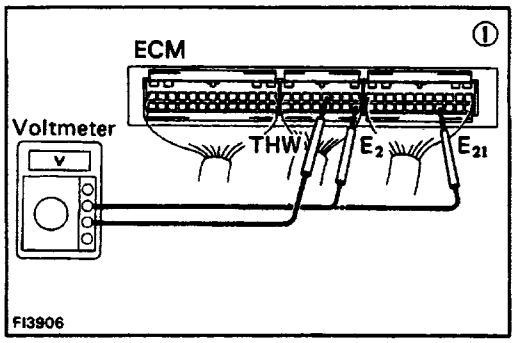
↓

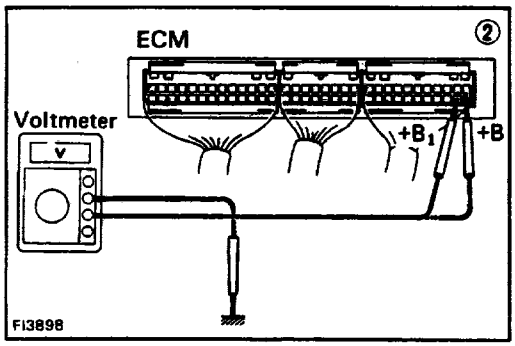
Try another ECM.

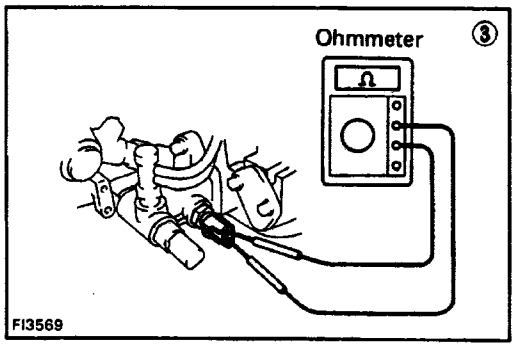
V01837

No.	Terminal	Trouble	Condition		STD Voltage
4	THW – E_2 (E_{21})	No Voltage	Ignition switch ON	Coolant temperature 80°C (176°F)	0.2 – 1.0 V









(1) There is no voltage between ECM terminals THW and EZ (E21–**(IG SW ON)**

(2) Check that there is voltage between EC M terminal + B 1 (+ B1) and body ground. **(IG SW ON)**

OK → Check wiring between ECM terminal E, and body ground.

NO → Refer to No. 1.

OK → (3) Check engine coolant temp. sensor.

BAD → Repair or replace.

BAD → Replace engine coolant temp. sensor.

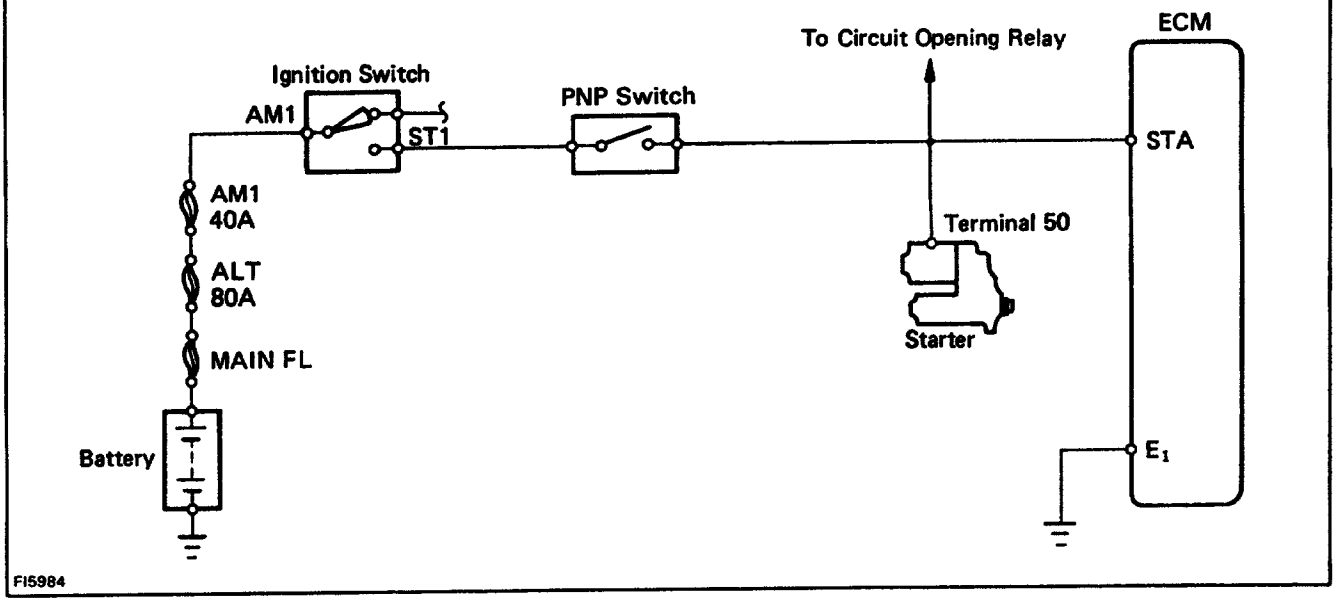
OK → Check wiring between ECM and engine coolant temp. sensor.

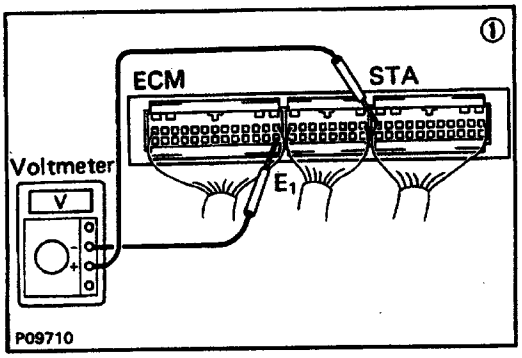
OK → Try another ECM.

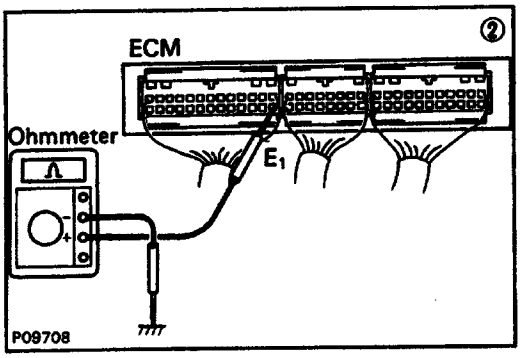
BAD → Repair or replace.

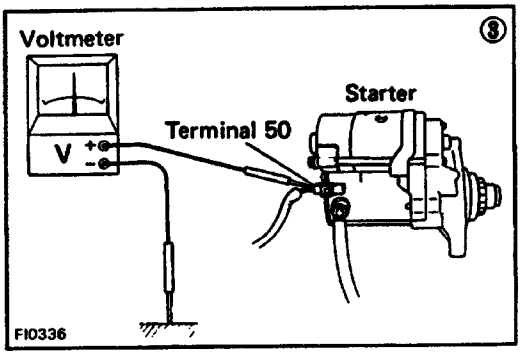
V01838

No.	Terminals	Trouble	Condition	STD Voltage
5	STA – E_1	No voltage	Ignition switch START position	6 – 12 V







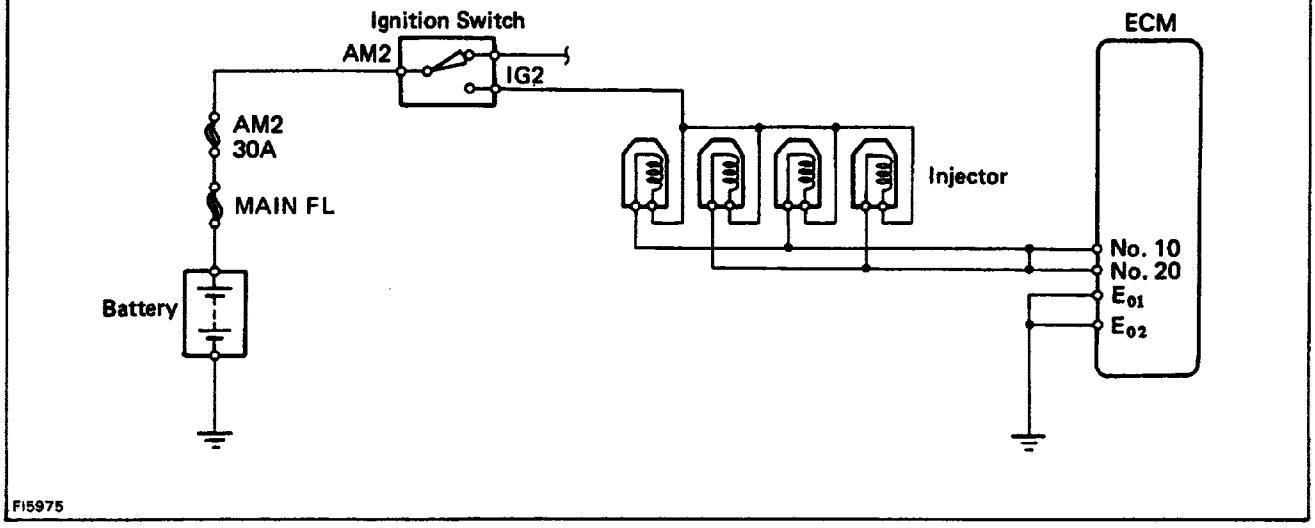


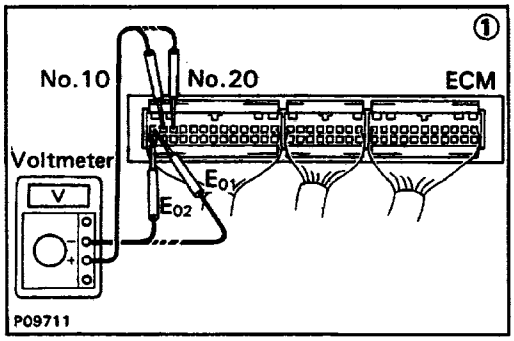
(1) There is no voltage between ECM terminals STA and E1 **(IG SW START)**

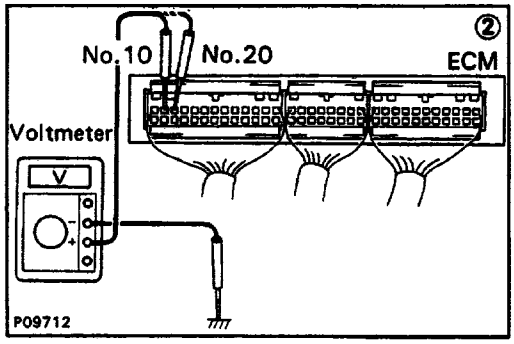
Check starter operation.
- OK → Check wiring between ECM and ignition switch terminal ST.
  - OK → (2) Check wiring between ECM terminal E, and body ground.
    - OK → Try another ECM.
    - BAD → Repair or replace.
  - BAD → Repair or replace.
- BAD → Check fusible link, battery, wiring and ignition switch.
  - BAD → Repair or replace.
  - OK → (3) Cheek that there is voltage at terminal 50 of starter. **(IG SW ST) STD Voltage: 6 – 12 v**
    - OK → Check starter. (See page ST–4)
    - NO → Check wiring between ignition switch ST1 terminal and starter terminal 50.

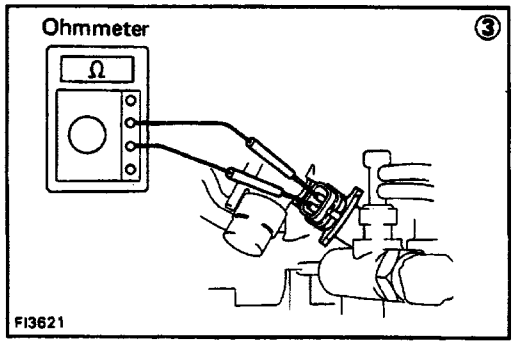
V01839

No.	Terminals	Trouble	Condition	STD voltage
6	No. 10 – E_{01} No. 20 – E_{02}	No voltage	Ignition switch ON	9 – 14 V









(1) There is no voltage between ECM terminals No. 10 and/or No. 20 and E0, and/or E02– **(IG SW ON)**

↓

(2) Check that there is voltage between ECM terminal No. 10 and/or No. 20 and body ground.

NO → **OK**

OK → Check wiring between ECM terminal Eo, and/or E02 and body ground.

OK → Try another ECM.

BAD → Repair or replace.

NO → Check fusible link and ignition switch.

BAD → Repair or replace.

OK →

(3) Check resistance of magnetic coil in each injector.
STD resistance: 13.4 – 14.2Ω

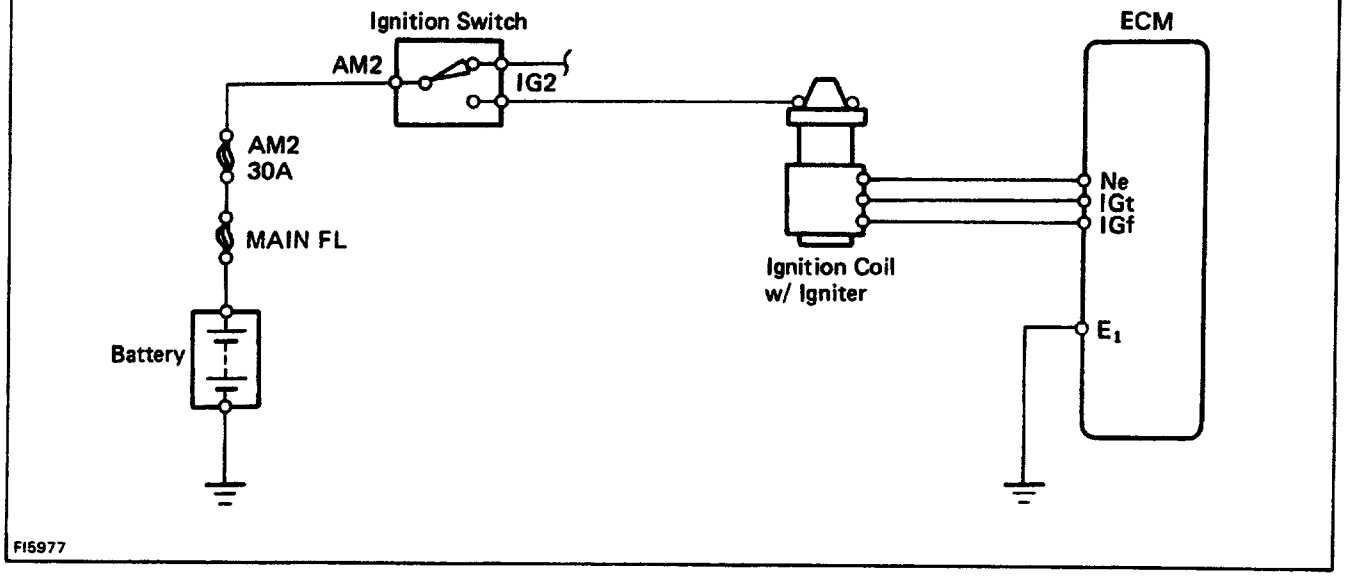
NO → Replace injector.

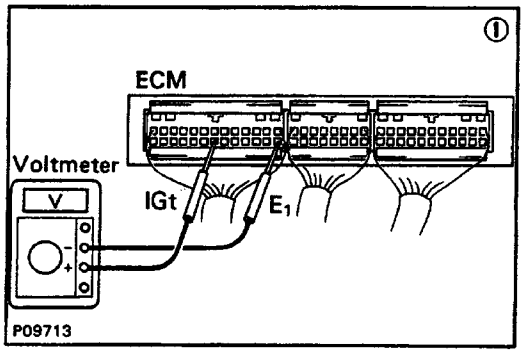
OK → Check wiring between ECM terminal No. 10 and/or No. 20 and battery.

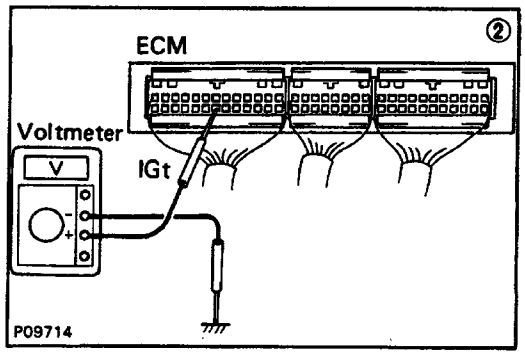
BAD → Repair or replace.

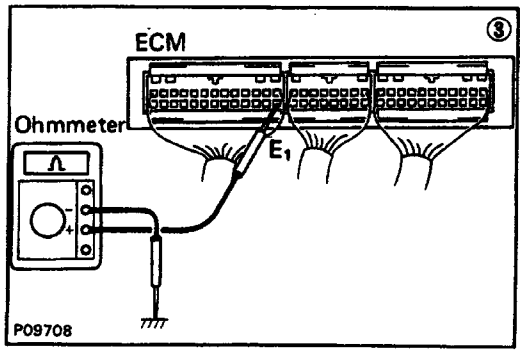
V01840

No.	Terminals	Trouble	Condition	STD Vol tage
7	IGt – E_1	No voltage	Idling	0.7 – 1.0 V









(1) There is no voltage between ECM terminals IGt and E1 **(idling)**

(2) Check that there is voltage between ECM terminal IGt and body ground. **(Idling)**

NO

OK

(3) Check wiring between ECM terminal E1 and body ground.

BAD

Refer to No. 1. — BAD → Repair or replace.

OK

Check wiring between igniter and distributor. — BAD → Repair or replace.

OK

Check distributor. — BAD → Replace.

OK

Check wiring between ECM and igniter. — BAD → Repair or replace.

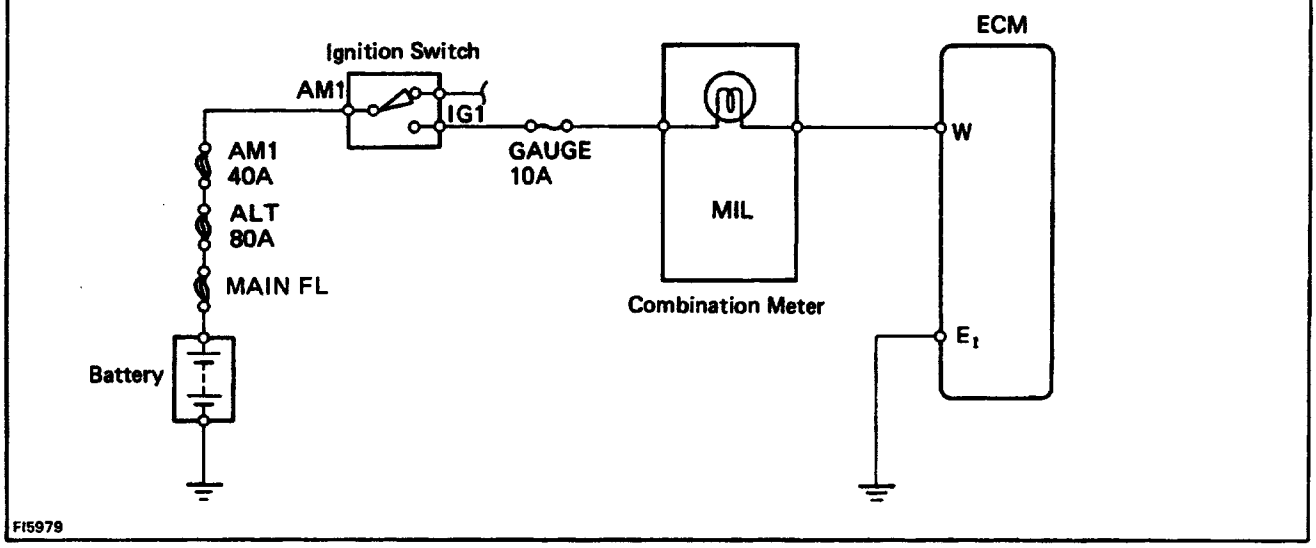
OK

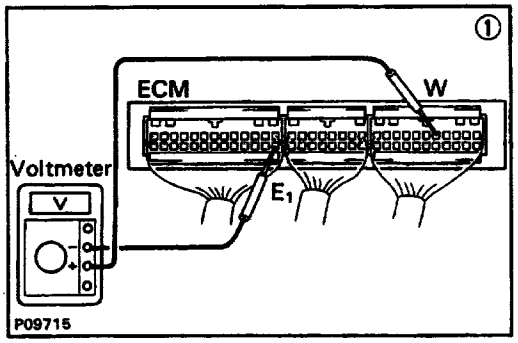
Check igniter. — BAD → Repair or replace.

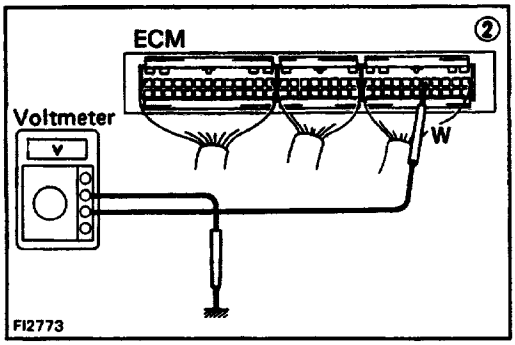
OK

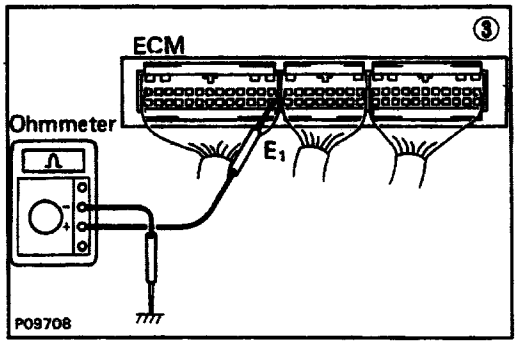
Try another ECM.

No.	Terminal:	Trouble	Condition	STD Voltage
8	W – E_1	No voltage	No trouble (MIL off) and engine running	9 – 14 V









(1) There is no voltage between ECM terminals W and E1 **(idling)**

(2) Check that there is voltage between ECM terminal W and body ground.

NO — **OK**

(3) Check wiring between ECM terminal E, and body ground.

OK — Try another ECM.

BAD — Repair or replace.

Check GAUGE fuse (10 A) and MIL.

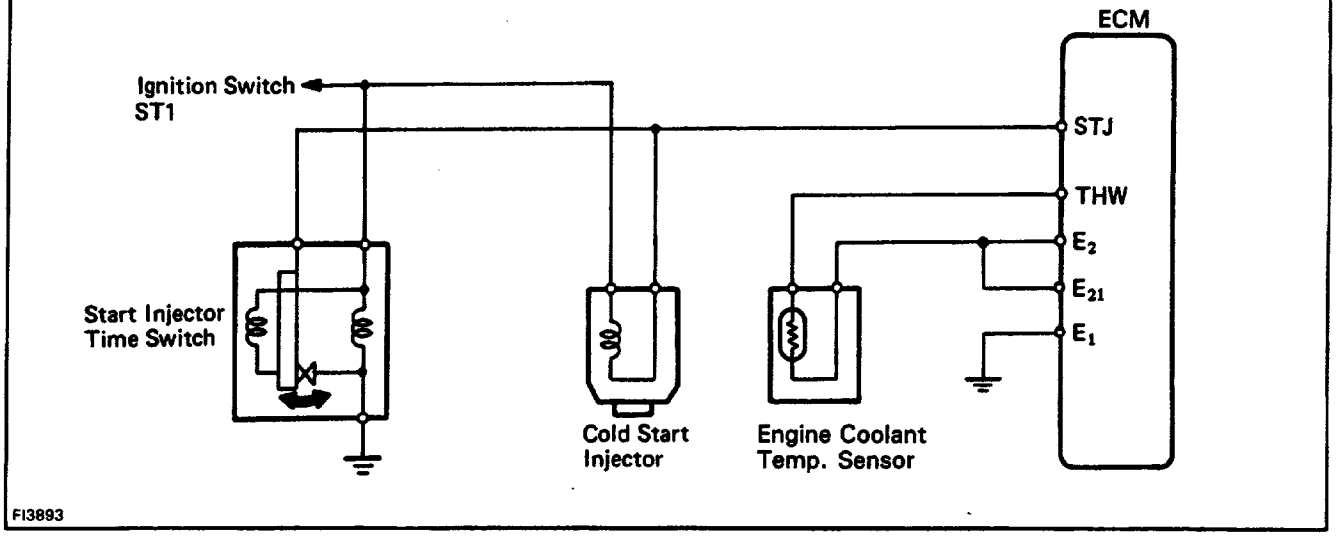
OK

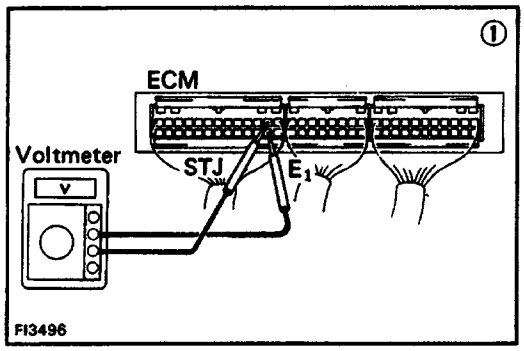
BAD — Repair or replace.

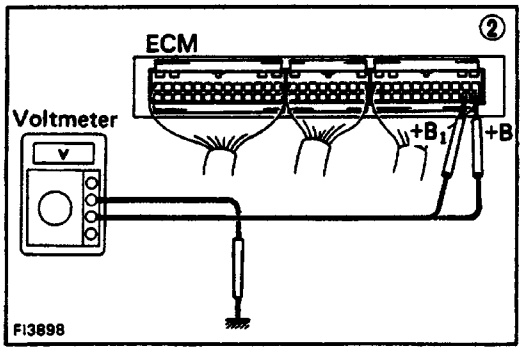
Fuse blows again
ECM BAD

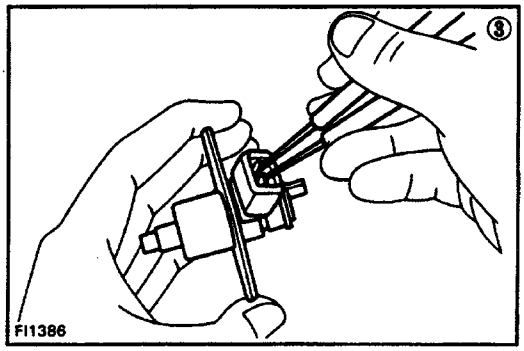
Check wiring between ECM terminal W and fuse. → Repair or replace.

No.	Terminal	Trouble	Condition		STD Voltage
9	STJ – E_1	No voltage	Ignition switch START position	Coolant temperature 80°C (176°F)	6 – 12 V









(1) There is no voltage between ECM terminals STJ and E1 **(IG SW START)**

↓

(2) Check that there is voltage between ECM terminal + B (+ B1) and body ground. **(IG SW ON)**

OK → ↓ | **NO** → Refer to No. 1.

↓

(3) Check cold start injector.

BAD → Replace cold start injector. → ↓

OK → Check wiring between ECM and cold start injector.

OK → ↓ | BAD → Repair or replace wiring.

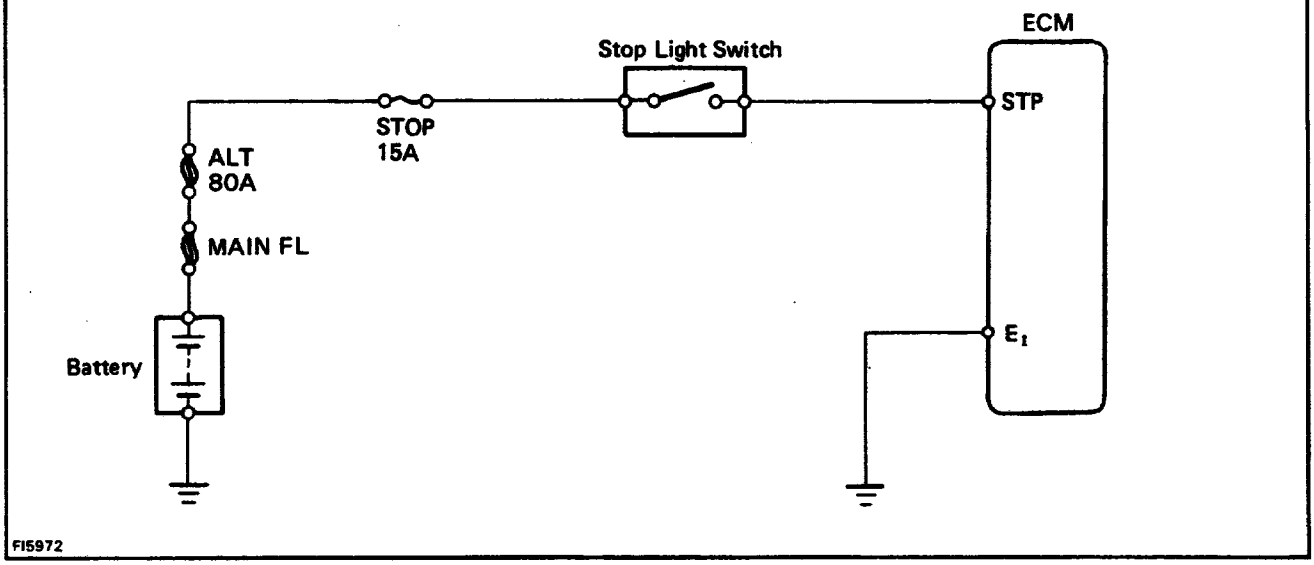
Check wiring between ECM terminal E, and body ground.

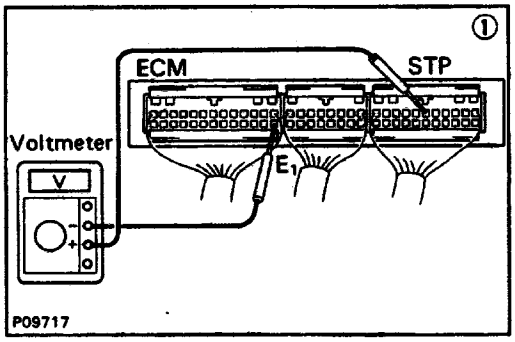
↓

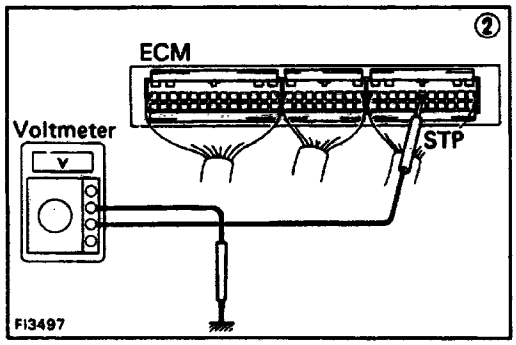
OK → Try another ECM.

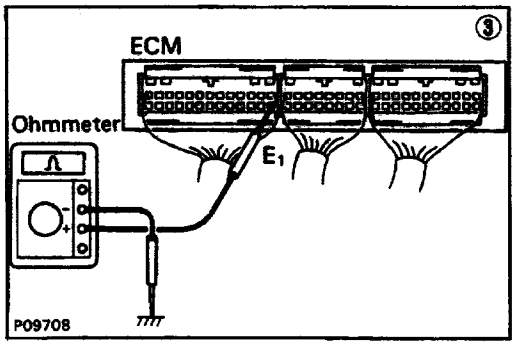
V01843

No.	Terminals	Trouble	Condition	STD Voltage
10	STP – E_1	No voltage	Stop light switch ON	7.5 – 14 V









(1) There is no voltage between ECM terminals STP and E1.

(2) Check that there is voltage between ECM terminal STP and body ground when the brake pedal .is depressed.

NO — OK

(3) Check wiring between ECM terminal .E, and body ground.

OK — BAD

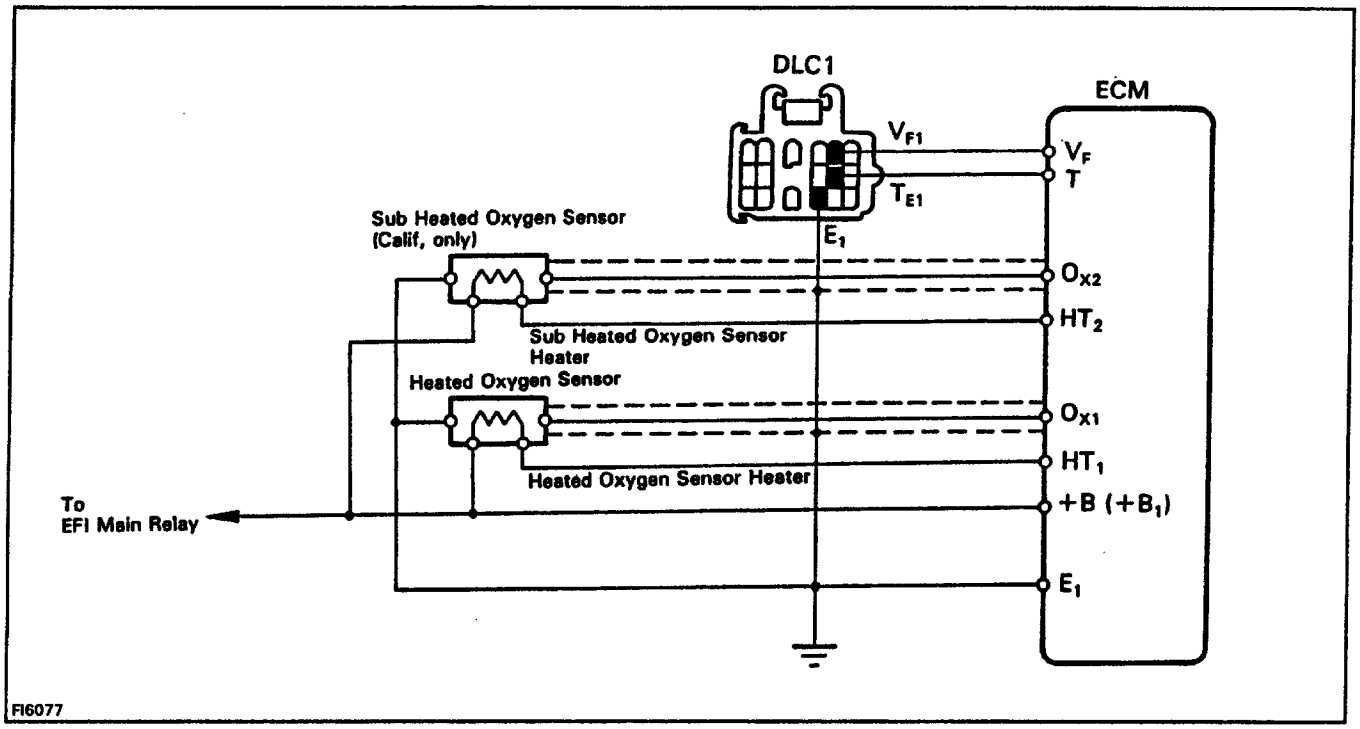
Try another ECM.

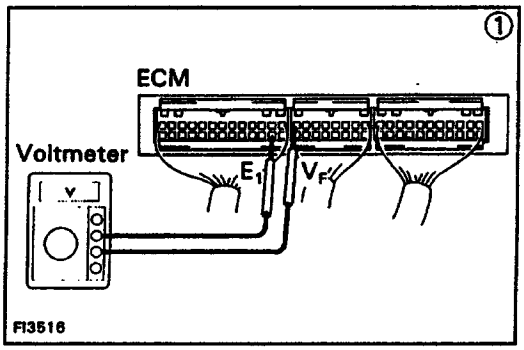
Repair or replace.

Check STOP fuse t 15A) and stop light switch. — BAD — Repair or replace.

OK

Check wiring between ECM terminal STP and battery. — BAD — Repair or replace.





(1) There is no voltage between ECM terminals VF and E1

↓

Check that there is voltage between ECM terminal VF and body ground.

NO → ↓

OK →

Check wiring between ECM terminal E, and body ground.

OK → Try another ECM.

BAD → Repair or replace.

↓

Is air leaking into air induction system?

YES → Repair air leak.

↓ NO

Check spark plugs.

BAD → Repair or replace.

↓ OK

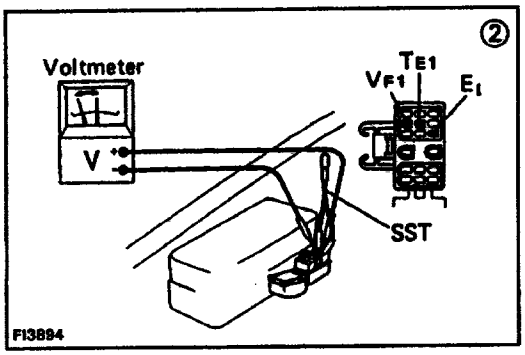
Check distributor and ignition system.

BAD → Repair or replace.

↓ OK

CONTINUED ON PAGE
EG1–160

V03955



CONTINUED FROM PAGE
EG1–159

OK

Check fuel pressure. — BAD → Repair or replace.

OK

Check injector. — BAD → Repair or replace.

OK

Check cold start injector.' — BAD → Repair or Replace.

OK

Check volume sir fiow meter. — 6AD → Repair or replace.

OK

(1) Check operation of heater ox en sensor — OK → System normal.

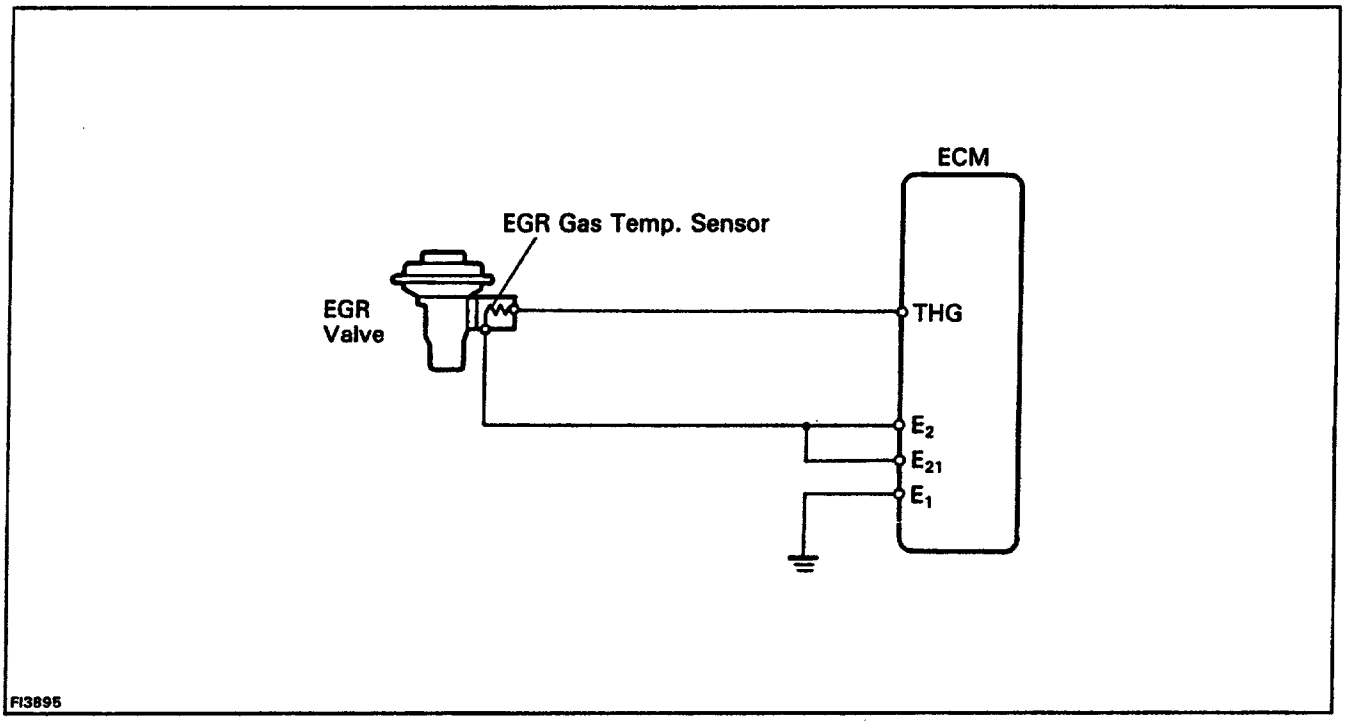
6AD

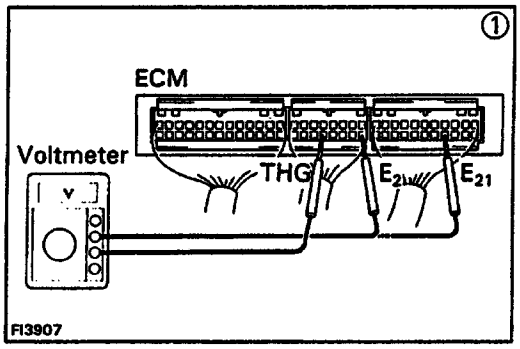
Check wiring between heated oxy–gen sensor and ECM connector. — BAD → Repair wiring.

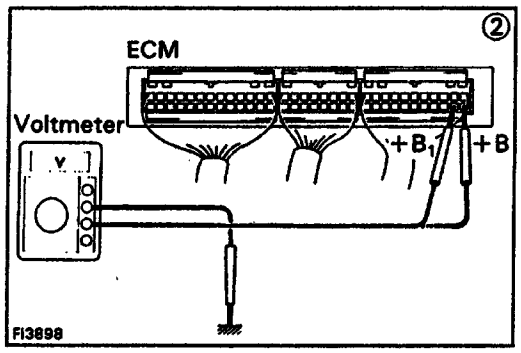
OK

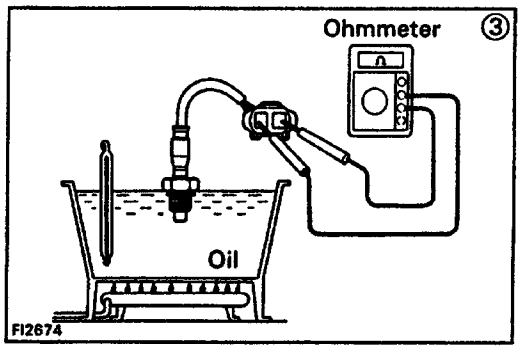
Replace heated oxygen sensor.

*Rich malfunction only









(1)There is no voltage between ECM terminals THG and Et (E21) (Engine running at 2,000 rpm)

(2) Check that there is voltage between ECM terminal +B (+B1) and body ground. **(IG SW ON)**

OK → Check wiring between ECM terminal E1 and body ground.

NO → Refer to No. 1 .

OK → Check EGR system.

BAD → Repair or replace.

OK → (3) Check EGR gas temp. sensor.

BAD → Replace EGR gas temp. sensor.

OK → Check wiring between ECM and EGR gas temp. sensor.

OK → Try another ECM.

BAD → Repair or replace.

BAD → Repair or replace.

V03952